

What Mizo reveals about the Architecture of D-Expressions

Abstract

This paper develops a unified analysis of definite and demonstrative descriptions, arguing that reference is decomposable into a small set of semantically independent building blocks: uniqueness in a restricted situation, an anaphoric index, a deictic relation, and anti-uniqueness in a larger situation. Drawing on original fieldwork data from Mizo, I show that cross-linguistic variation in definite and demonstrative expressions (henceforth D-expressions) reflects different ways of distributing these components across the nominal spine. A key finding is that what distinguishes a demonstrative description is not deictic value per se, but an anti-uniqueness presupposition. The Mizo data make this transparent: several determiner morphemes encode deictic properties, yet without the anti-uniqueness component, such determiners yield definite rather than demonstrative descriptions. In Mizo, D-expressions appear in two morphological configurations, NP-D and D-NP-D, which diverge in precisely two distributional properties: compatibility with globally unique descriptions (such as *the sun*) and the ability to be used contrastively (as in *X while Y*). I argue that this split reduces to anti-uniqueness: its presence renders an expression incompatible with globally unique descriptions and licenses contrastive use, while its absence has the opposite effect (Robinson 2005; Dayal and Jiang 2022; Dayal *in prep*). Within the proposed framework, the D-NP-D configuration is maximally specified, combining anti-uniqueness (contributed by the pre-nominal morpheme) with the remaining components supplied by the post-nominal D; it is this combination, rather than the demonstrative category as such, that is semantically maximal. The analysis extends cross-linguistically to Akan and Bangla, showing that while all three languages share the same division of labor between reference, indexicality, deixis, and anti-uniqueness, they differ in how these components are morphosyntactically realized. The proposed decomposition also bears on the distinction between *this* and *that* in English (Elbourne 2008): I tentatively explore the possibility that the two differ in whether an anaphoric index is encoded, with *this* patterning with Mizo deictic expressions in lacking one, though a full treatment is left for future work. The paper advances two central claims: anti-uniqueness is an independent semantic ingredient rather than a pragmatic by-product of deixis or anaphora, and the traditional opposition between definites and demonstratives reflects different combinations of independently available reference-building mechanisms.

1 Introduction

1.1 Building blocks of definiteness: *uniqueness* and *familiarity*

Starting with Frege (1892), Russell (1905), and Strawson (1950), questions concerning the nature of definiteness and its expression in language have occupied formal semanticists for decades. The

notion of definiteness is standardly characterized by two key properties: *uniqueness* and *familiarity*. The question of whether languages are attuned to the nuance between these two has attracted considerable attention in recent literature (Schwarz 2009; Arkoh and Matthewson 2013; Biswas 2012; Jenks 2018; Coppock 2022; Ahn 2019; Dayal 2026; Dayal *in prep*, a.o.). Schwarz’s (2009) work laid the groundwork, showing that the distinction surfaces morphologically in German and Fering through a contrast between *weak* and *strong* definite forms. In German, a preposition and a definite article can contract in non-anaphoric contexts, as with *vom* ‘from-the_{weak}’ in (1a), but not in anaphoric ones, as in (1b), where only the uncontracted form *von dem* ‘from the_{strong}’ is licensed. Several consequent accounts of unrelated languages have adopted this dichotomy.

- (1) a. *Der Apfel fällt nicht weit vom* / *#von dem* *Stamm.*
the apple falls not far from-the_{weak} / from the_{strong} (tree-)trunk
‘The apple doesn’t fall far from the trunk of the tree.’ (saying)
- b. *Hans hat einen Schriftsteller und einen Politiker interviewt. Er hat #vom*
Hans has a writer and a politician interviewed. He has from-the_{weak}
/ *von dem* *Politiker keine interessanten Antworten bekommen.*
/ from the_{strong} politician no interesting answers gotten
‘Hans interviewed a writer and a politician. He didn’t get any interesting answers from the politician.’
(German; Schwarz 2009)

The same underlying question of how the semantic components of definiteness are distributed across morphological forms extends to the distinction between *definite* and *demonstrative* descriptions, a further nuance equally debated in the literature. In Kaplan’s (1989) classical theory, demonstratives establish reference differently from definite descriptions. Their reference is (1) *direct*: the referent enters the proposition as a bare individual, unmediated by any descriptive content, with a pointing gesture fixing which individual that is; and (2) *rigid*, immune to intensional operators, as illustrated by the referent of *this man I’m pointing at* in (2b) below.; and (2) *rigid*, immune to intensional operators, as illustrated by the referent of *this man I’m pointing at* in (2b) below. Both properties follow naturally from the fact that demonstratives encode deixis: the object being pointed at is the referent, and it remains so as long as the pointing persists.

- (2) *Context: A Spanish and a French footballer are being interviewed after a match. The French footballer is sitting to the right of the Spanish footballer. The person next to you points at the French footballer and says:*
- a. If they exchanged seats, *the* man I’m pointing at would be from Spain.
b. If they exchanged seats, *this* man I’m pointing at would be from Spain.
(adapted, Kaplan 1989:517)

In fact, deixis has been taken to play the instrumental role in setting demonstratives apart from definite descriptions, though subsequent work has shown that the descriptive content matters too. Along this line, demonstratives have been argued to be distinctive in their ability to combine gestural information (pointing) with linguistic information (description), functioning as

a *modality linker* (Ebert and Ebert 2014; Ahn 2022; Ahn 2025). Yet demonstratives are not always used deictically. In uses not supplemented by co-speech pointing, as in (3) below, they pattern closely with definite descriptions, and their referent is retrieved through the description rather than through direct acquaintance. Moreover, as (4) illustrates, the referent can covary across contexts, raising the question of what distinguishes demonstratives from definites in such cases (King 2001; Roberts 2002; Elbourne 2008).

(3) A student_{*i*} walked in. *That student_{*i*}* had recently joined the school.

(4) That professor who brought in the biggest grant in each division will be honored.
(King 2001: 3-10)

Some have maintained that deixis remains significant, arguing that demonstratives encode both an anaphoric index and a deictic relation, the two together fixing the referent (Elbourne 2008), while others have located the demonstrative-definite distinction in an *anti-uniqueness* component: the referent is unique in the relevant situation of use, but there exist at least two individuals satisfying the description in some larger situation (Robinson 2005; Nowak 2019; Dayal and Jiang 2022), exemplified in (5) below.

- (5) a. The author of Waverley also wrote Ivanhoe.
b. #That author of Waverley also wrote Ivanhoe.

(Nowak 2019: 2867)

In this paper, I argue that these distinctions between unique and familiar definites on the one hand, and between definites and demonstratives on the other, are not separate puzzles but facets of the same underlying inquiry into definiteness. Definiteness, I propose, comprises a small set of distinct semantic pieces, combinations of which give rise to a range of expressions (unique definites, familiar definites, and demonstratives) which I collectively refer to as *D-expressions* henceforth. Drawing on novel fieldwork data from Mizo, a Tibeto-Burman language that deploys dedicated morphemes for deixis, anaphora, and anti-uniqueness (components typically fused within a single determiner in languages like English), I show that the internal architecture of D-expressions can be made unusually transparent. Specifically, I propose that the relevant pieces are: uniqueness in a restricted situation (U_{S_r}), an anaphoric index ($\mathcal{I}$), a deictic relation ($\mathcal{R}$), and anti-uniqueness in a larger situation (AU_{S_l}). Mizo is particularly revealing in that it deploys dedicated D-morphemes for deixis and anaphora, a distinction that holds for both definite and demonstrative forms, expressed in NP-D and D-NP-D configurations respectively. What sets the two apart is not deixis, which is present in both, but the encoding of anti-uniqueness in the D-NP-D form. Crucially, deixis is *neither sufficient nor necessary* for demonstrative reference; while it does enforce rigid reference (à la Kaplan), it does not suffice for contrastive uses such as *that dog_{→A} is running, while that dog_{→B} is sleeping*. For this, anti-uniqueness is essential, lending crosslinguistic support to Robinson (2005), Nowak (2019), Dayal and Jiang (2022), and Dayal (in prep). The result is a four-way typology of D-expressions in Mizo: deictic definites and anaphoric definites in the NP-D configuration, and deictic demonstratives and anaphoric demonstratives in the D-NP-D configuration. The implications extend beyond Mizo, bearing on the structure of demonstrative systems crosslinguistically, including English.

2 Probing Definiteness: The Case of Mizo

This section presents the empirical landscape of definite and demonstrative expressions in Mizo (Tibeto-Burman), drawing on data collected during fieldwork conducted in Mizoram in Fall 2024.

2.1 Situating Mizo within Formal Approaches to Definiteness

As discussed above, demonstratives are commonly analyzed as lexically encoding deictic features, most often a proximal–distal distinction, as in English *this* and *that*, with pointing gestures directing the interlocutors’ attention to a referent relative to the speaker’s orientation. On Kaplan’s account, this deictic component is captured as a directing intention, and as Ahn (2022) and Ahn (2025) argue, demonstratives crucially involve NP-internal deictic composition, a property not shared by definite descriptions. *Deixis* and *demonstrative descriptions* are therefore taken to be tightly linked across these accounts.

When we turn to Mizo, however, we find that these two notions can come apart. A DP such as *lekhabu-saw*, ‘book-DIST’, which may be glossed as *the book that is over there*, encodes a distal deictic value in that it is incompatible with objects in the speaker’s proximity, yet fails to pattern like a genuine demonstrative. First, it cannot be used pronominally: unlike English *that*, Mizo *saw* requires overt NP content and cannot head a DP on its own. Second, it does not support contrastive uses even when assisted with pointing. Lastly, it is compatible with globally unique referents such as the sun (6b).

- (6) a. #*lekhabu-saw* (pointing at book A) ka duh, *lekhabu-saw* (pointing at book B) ka
 book-DIST 1.SG want, book-DIST 1.SG
 duh lo
 want NEG
 Intended: ‘I want that book (pointing at book A), I don’t want that book (pointing at book B).’
 b. *ni-saw en teh!*
 sun-DIST look IMP
 ‘Look at the sun over there.’ (Mizo)

These properties suggest that, in Mizo, deictic morphology does not by itself give rise to demonstrative reference, motivating a re-examination of how deixis, definiteness, and demonstratives are related. Specifically, I argue, following Dayal and Jiang (2022) and Dayal (in prep), that what distinguishes a D-NP-D from an NP-D is the inclusion of an anti-uniqueness condition in the denotation of D-NP-D, which effectively results in D-NP-D patterning as a demonstrative rather than a definite. To elucidate, a felicitous use of a demonstrative, such as *that F*, in a situation *s* requires *F* to be satisfied by more than one individual in a situation larger than *s*. A key property distinguishing definites from demonstratives is that demonstratives permit non-uniqueness (Robinson 2005; Dayal in prep), which effectively licenses contrastive use, as in (7a) below. In fact, demonstratives *require* non-uniqueness, as made evident in constructions like (7b) and (7c).

- (7) a. That/ #The dog is sleeping, that/#the dog is running around. (in a context involving two dogs)
- b. The/ #That sun is shining bright.
- c. The match was interesting. The/ #That referee was unfair.

The infelicity in (7b) follows from the fact that, even in the largest relevant situation, the predicate *sun* is satisfied by a unique individual – our sun^{1,2}.

In cases like (7c), although the description *referee* is not globally unique in isolation, it becomes unique in the restricted situation determined by the previously mentioned *match*: there is exactly one individual of whom the predicate *referee* is true – the referee of that match – and thus the anti-uniqueness requirement of the demonstrative cannot be met.

The anti-uniqueness approach has been argued to be too strong. Blumberg (2020) shows that overt rejection of the anti-uniqueness presupposition does not preclude the felicitous use of demonstrative descriptions, as in (10a). However, Saha (2025) has argued that this results from the speaker’s lack of knowledge regarding the referent’s uniqueness. Thus, a minor modification of (10a), illustrated in (10b), brings the known uniqueness of the object into conflict with the anti-uniqueness requirement of the demonstrative.

- (10) a. I don't know if there are any other cars in this showroom, but [that car]_→ looks expensive.
(Blumberg 2020:49)
- b. #I know there aren't any other cars in this showroom, but [that car]_→ looks expensive.
(Saha 2025: 1387)

I further contend that the restriction for demonstratives is not the uniqueness of the object itself, but the description under which it is introduced. The sun, qua round thing, is not globally unique: there are many round objects in the world, and we can point to the sun as one of them, as in (11a). By contrast, the sun qua sun is unique. This explains the contrast between (11b) and (11a).

- (11) a. Can you believe that **round thing** is the source of energy for all human life?
b. #Can you believe that **sun** is the source of energy for all human life?

In what follows, I lay out the operative empirical facts in Mizo before turning to the analysis.

¹*That sun is*, of course, acceptable once the relevant situation is extended beyond our solar system. Thus (8) is felicitous in a context involving astronauts observing multiple suns across different solar systems or galaxies.

- (8) *That sun is a thousand years older than ours.*

²Demonstratives *this* and *that* also have “emotive” uses. These signal affective or interactional closeness between the discourse participants (Lakoff 1974), and pattern closely to how uncontroversially exclamative expressions are evaluated (Potts and Schwarz 2010). As illustrated in (9) below, *that sun* is felicitous under an emotive use, in parallel to another uniquely denoting expression, such as a proper name, when used with demonstratives (see Dayal 2026: Chapter 2).

- (9) a. *That sun will give me a heat stroke!*
 b. *That Bondi should be jailed!*

2.2 Deictic and anaphoric definites and demonstratives in Mizo

A series of determiner morphemes are employed in definite and demonstrative descriptions: four encode different deictic properties, and two enable anaphoric links. These determiners are as illustrated in (12). When they appear post-nominally, in an NP-D configuration, the resulting expression is interpreted as a definite (13a), while demonstratives correspond to a circumnominal D-NP-D configuration (13b).

(12)

proximal	distal	above	below	anaphoric	anaphoric
<i>hi</i>	<i>saw</i>	<i>khi</i>	<i>khu</i>	<i>chu</i>	<i>kha</i> ³

- (13) a. *ui hi/ ui chu*
dog PROX/ dog INDEX
‘The dog here/ The (pre-mentioned) dog.’
- b. *he ui hi/ chu ui chu*
PROX dog PROX/ INDEX dog INDEX
‘This dog/ That (pre-mentioned) dog.’
(Mizo)

There are two ways in which bare nouns function as arguments in Mizo; on one use, they get a number-neutral interpretation (strictly as non-specific indefinites). In this use, Mizo bare nouns parallel English bare plurals, as illustrated in (14–15) below. The NP-D and D-NP-D forms can only be interpreted as definites or demonstratives. As illustrated in (14a) and (15a), the bare noun is necessarily interpreted as a (number-neutral) non-specific indefinite, and therefore cannot be used for referring to the (maximal sum of) relevant guests. Conversely, NP-D/D-NP-D forms in (14b) and (15b) necessarily denote the (maximal sum of) relevant guests.

- (14) a. *mikhual chu/ #mikhual an chhuak*
guest INDEX/ guest 3.PL leave
dawn
MOD
Intended: ‘The (relevant) guests are leaving.’
- b. *#{mikhual chu}/ mikhual an*
guest INDEX/ guest 3.PL
chhuak dawn
leave MOD
Intended: ‘Guests are leaving.’ (Mizo)
- (15) a. *chu mikhual chu/ #mikhual*
INDEX guest INDEX/ guest
an chhuak dawn
3.PL leave MOD
Intended: ‘Those (relevant) guests are leaving.’
- b. *#{chu mikhual chu}/ mikhual an*
INDEX guest INDEX/ guest 3.PL
chhuak dawn
leave MOD
Intended: ‘Guests are leaving.’ (Mizo)

2.3 The definite-demonstrative dichotomy in Mizo

The two forms, NP-D and D-NP-D, pattern typically with definite and demonstrative descriptions respectively. This was tested using Dayal’s (2026) questionnaire on definites and indefinites in languages with bare noun arguments. As illustrated in (16b), NP-D forms are infelicitous in

³Given the limited data available for anaphoric *kha*, the discussion of anaphoric definites and demonstratives in this paper focuses on *chu*.

contexts involving more than one object with the relevant NP property. In (16b), the NP-D *does not pass* the contrastiveness test, which involves combining two predicates that are mutually incompatible – ones that cannot be true of a single individual simultaneously. Patterning with demonstratives, the D-NP-D form in (16a) can be used felicitously.

- (16) a. *he ui hi a mu leh he ui hi a tlan*
 PROX dog PROX 3.SG sleep and PROX dog PROX 3.SG run

‘This dog is sleeping and this dog is running.’

- b. *#ui hi a mu leh ui hi a tlan*
 dog PROX 3.SG sleep and dog PROX 3.SG run

Intended: ‘This dog is sleeping and this dog is running.’

(Mizo)

It is important to note that NP-D forms cannot be used contrastively even though they carry deictic information, and even if the deictic values differ – proximal vs. distal, for instance (17b). Similar constructions in languages like English, on the other hand, are felicitous (17a). This indicates that the deictic morpheme does not function simply as a modifier (as in English), but necessitates uniqueness of the NP property in the utterance context. Moreover, the NP-D form is not felicitous when used even non-contrastively in a context involving more than one individual that satisfies the NP property, as in (18).

- (17) a. I want the jacket over here not the jacket over there.

(English)

- b. *#jacket hi ka duh, jacket saw ka duh lo*
 jacket PROX 1.SG want, jacket DIST 1.SG want NEG

Intended: ‘I want the jacket over here not the jacket over there.’

(Mizo)

- (18) *Context: spoken by a customer at a shop, pointing at one rose among several in a bunch...*

#rose hi ka lo lei
 rose DET:PROX 1SG MOD buy

Intended: ‘I’ll buy this rose.’

(Mizo)

Conversely, D-NP-D forms are infelicitous when used with globally unique objects such as the sun or the prime minister of the country, while NP-D forms are not. Moreover, D-NP-D forms fail to be used for entities that are functionally a part of something previously mentioned; e.g., the referee of a football match, as in (20).

- (19) a. *Context: The speaker and addressee are hiking and the sun looks magnificent..*

#{saw ni saw}/ ni saw en teh
 DIST SUN DIST/ SUN DIST look IMP

Intended: ‘Look at the sun (over there)!’

- b. *Context: The speaker and addressee are at a press conference where the Prime Minister is present somewhere a little far from them...*

#{**saw** PM **saw**}/ PM **saw** a upa hmel e
 DIST PM DIST/ PM DIST 3.SG old look MOD?

Intended: ‘The PM, who’s standing over there, looks old.’ (Mizo)

- (20) *Context: The speaker is reporting something to the addressee that she heard from her friend Dee.*

She says: Dee was watching football,...

#{**chu** referee **chu-an**}/ referee **chu-an** tran lam a nei
 INDEX referee INDEX-ERG/ referee INDEX-ERG support side 3.SG have

‘The referee was partial.’ (Mizo)

As is common for definites and demonstratives, there are contexts where the two forms, NP-D and D-NP-D, behave similarly. One such context is an immediate situation uniqueness use, where the referent is unique in the restricted situation made relevant by the utterance context, illustrated in (21), but the cardinality of the set of entities satisfying the NP property exceeds 1 in the largest relevant situation. Take the context of utterance to be one where there is a single car which happens to be new: both *the new car* and *that new car* are felicitous. Compare this to a context where people are talking about the sun: #*that sun* is infelicitous while *the sun* is not. This is because *sun* denotes a globally unique entity, at least insofar as we restrict the domain to our solar system, and thus the anti-uniqueness requirement of the demonstrative cannot be satisfied.

- (21) *Context: Talking about a new car that someone bought...*

he car **hi**/ car **hi** thar a ni
 PROX car PROX/ car PROX new 3.SG is

‘This/The car is new.’ (Mizo)

The second context where the two forms behave similarly is anaphoric use, illustrated in (22a); however, D-NP-D forms are degraded in cases like (22b), where the expression is used to refer to an entity previously introduced in the discourse alongside another entity. The distribution of labor between the two forms thus appears consistent with the known differences between definite descriptions and demonstrative constructions, summarized in Table 1 below.

- (22) a. *Context: The speaker is telling the addressee about what she saw on the road.*
She says: We saw a huge cow,...

chu bawng **chu**/ bawng **chu** a lo mu
 DET:AN COW DET:AN/ COW DET:AN 3.SG MOD sleep

‘That/ The cow was sleeping.’ (Mizo)

- b. *Context: The speaker is reporting a scenario to the addressee*
She says: A boy and a girl came into the room...

#{**chu** mipa.naupang **chu-an**}/ mipa.naupang **chu-an** cap a ha
 DET:AN boy DET:AN-ERG/ boy DET:AN-ERG cap 3.SG wear

‘A boy and a girl came into the room. #{That}/ The boy was wearing a cap.’ (Mizo)

Condition	Environment	NP-D	D-NP-D
Similar	Situationally unique	✓	✓
	Anaphoric	✓	✓
Different	Globally unique	✓	✗
	Non-unique	✗	✓
	Functional	✓	✗
	Contrastive	✗	✓

Table 1: Distribution of NP-D and D-NP-D in Mizo

As established above, deixis has long been associated with demonstratives, yet its role in definite descriptions has received comparatively little attention⁴. Numerous accounts of demonstratives ultimately locate the definite-demonstrative contrast solely in deixis, claiming it is the only crosslinguistically robust distinction (Levinson 2004, Diessel 2006, Hughes 2020). The Mizo data, however, point in a different direction.

3 A (de)composable D-expressions account

I propose that both definite and demonstrative descriptions can be subsumed under the category of D-expressions, which are employed for reference crosslinguistically. I argue that there are incrementally informative semantic blocks that build D-expressions; definites *and* demonstra-

⁴Robinson (2005) uses the term ‘deictic definite’ in a broader sense, referring to cases in which a definite determiner can be used to point to an intended referent that is present in the utterance context. She uses this to draw a contrast between English and French: in English, the definite determiner may be used to identify an entity present in the utterance context, whereas French disallows such usage, as illustrated in (23). Crucially, my use of the term ‘deictic definite’ is narrower: it refers to Mizo definite determiners that *lexically encode* a deictic feature – proximal (*hi*), distal (*saw*), above (*khi*), below (*khu*)

- (23) *Context: two friends touring the countryside enter a new village, and head to the town square, where they find a baobab tree. In the baobab is sitting a monkey. One friend says to the other:*

- a. The monkey is giving you a funny look!

- b. Ce singe/ #Le singe te regarde d’un drôle œil
 DEM.SG.M monkey/ DEF.SG.M monkey 2.SG.REFL look-at.3.SG of-INDEF.SG funny eye

Intended: ‘The/That monkey is giving you a funny look!’ (French; cited by Robinson 2005: 40)

tives. Four informative blocks constitute this class: uniqueness in the restricted situation (U_{S_r}), an anaphoric index ($\mathcal{I}$), a deictic property ($\mathcal{R}$), and anti uniqueness in the larger situation (AU_{S_l}). The particular subset of these properties associated with a D-expression determines whether it patterns as a demonstrative description or as a definite description. Languages lexicalize these properties differently, but a language could, in principle, morphologically spell out each one. The minimal and necessary condition for definiteness is U_{S_r} , and the rest are built upon it. While $\mathcal{I}$ and $\mathcal{R}$ make at-issue contributions, AU_{S_l} only contributes a presupposition. The general schemata for D-morphemes are provided in (24) below.

- (24) a. $\llbracket D:U_{S_r} \rrbracket^{c,g} = \lambda P \lambda s_r. \iota x [P(x)(s_r)]$ if $\exists! x [P(x)(s_r)]$, undefined otherwise.
 where s_r is the restricted situation provided by c, in which uniqueness is evaluated
- b. $\llbracket D:U_{S_r} + \mathcal{I} \rrbracket^{c,g} = \lambda P \lambda s_r. \iota x [P(x)(s_r) \wedge g(i)(s_r) = x]$ if $\exists! x [P(x)(s_r) \wedge g(i)(s_r) = x]$, undefined otherwise.
 where s_r is the restricted situation provided by c and i is an index
- c. $\llbracket D:U_{S_r} + \mathcal{R} \rrbracket^{c,g} = \lambda P \lambda s_r. \iota x [P(x)(s_r) \wedge \mathcal{R}(x, \mathcal{I})(s_r)]$ if $\exists! x [P(x)(s_r) \wedge \mathcal{R}(x, \mathcal{I})(s_r)]$, undefined otherwise.
 where $\mathcal{R}$ is the deictic relation encoded in D and $\mathcal{I}$ is the speaker
- d. $\llbracket D:U_{S_r} + AU_{S_l} \rrbracket^{c,g} = \lambda P \lambda s_r. \iota x [P(x)(s_r)]$ if $\exists! x [P(x)(s_r) \wedge \exists s_l s_r \leq s_l |P(x)(s_l)| > 1]$, undefined otherwise.

Some D-expressions encode U_{S_r} together with $\mathcal{I}$, yielding strong definites. Others encode U_{S_r} plus $\mathcal{R}$, giving rise to deictic definites. The maximal D-expression encodes all four- U_{S_r} , $\mathcal{I}$, $\mathcal{R}$, AU_{S_l} , corresponding to *demonstratives*. The added layer of AU_{S_l} is what facilitates such D-expressions to be contrasted with each other, since it specifies that the cardinality of the NP set in a larger situation is more than 1. The argument s is saturated by the utterance context as the final step of evaluation.

3.1 Rigidity and Covariability: *deictic* vs. *anaphoric* D-expressions

Before turning to how these components combine to derive definites and demonstratives in Mizo, I first clarify the individual contributions of $D:U_{S_r} + \mathcal{R}$ and $D:U_{S_r} + \mathcal{I}$, that is, the distinction between *deictic* and *anaphoric* D-expressions. In many languages, this distinction is not transparent, since demonstratives often simultaneously encode both an index and a deictic property, as illustrated below in (25) for English *that*.

- (25) a. Josh bought a book_{*i*} last week. ***That book_{*i*}*** costs over \$1000!
 b. Josh bought ***that _{$\rightarrow$} book***

Mizo, however, morphologically separates these components. D-expressions encoding a deictic value, whether definite (NP-D) or demonstrative (D-NP-D), *cannot* be used anaphorically to refer to a previously mentioned discourse entity: the morphemes *hi*, *khi*, *khu*, and *saw* are

restricted to reference within the interlocutors’ perceptual field, as illustrated in (26a) below. These morphemes therefore instantiate the configuration $D:US_r + \mathcal{R}$.

(26) *Some guests visited us from Aizawl yesterday...*

- a. #mikhual-saw/ #saw mikhual-saw an chhuak ta
 guest-DIST/ DIST-guest-DIST 3.PL leave MOD

Intended: ‘The/those guests have left.’

- b. mikhual-chu/ chu mikhual-chu an chhuak ta
 guest-INDEX INDEX-guest-INDEX 3.PL leave MOD

‘The/Those guests have left.’

(Mizo)

In other words, the D-morphemes that assign deictic values to the referent, namely *hi*, *saw*, *khi*, and *khu*, do not encode an index, which in Mizo is exclusively realized by the morpheme *chu*. Since these morphemes exclusively encode a deictic value, to the exclusion of an index, definite expressions formed with such morphemes exhibit a range of properties commonly associated with demonstratives and argued to be *specific* to their deictic uses.⁵ I focus on two such diagnostics: *at-issue pointing* and *rigid designation*.

As briefly mentioned in §1, demonstratives have been argued to make a co-speech pointing gesture at-issue, while in the case of definites it remains not-at-issue (Ebert and Ebert 2014, Ebert, Ebert and Hörnig 2020). For example, the demonstrative description *that car*, when supplemented by pointing, introduces an at-issue commitment that the referent is identical to the object being pointed at. Ahn (2022) argues that this characterization makes the contribution of a demonstrative too restrictive, and proposes instead that definite descriptions allow for a mismatch between the demonstratum and the description. She illustrates with an example where the speaker utters (27) while pointing at a person.

(27) Cornelia bought the bottle_→

The resulting inference is not one of identity but of *relevance*: namely, that the individual being pointed at stands in a salient relation to the bottle being indicated. On her account, pointing contributes supplementary information when used with definite descriptions. By contrast, demonstratives incorporate pointing information as a property that composes noun-internally; in other words, pointing is restrictive for demonstratives but non-restrictive for definite descriptions. Elbourne’s (2008) treatment of demonstratives similarly treats deixis as an at-issue component of the demonstrative. Against this backdrop, I turn to the behavior of deictic definites

⁵At this stage, it is important to flag what distinguishes forms such as NP-D and D-NP-D, and why I have characterized the former as a definite expression and the latter as a demonstrative. Recall that a deictic D-expression is an expression headed by a D-morpheme that exclusively encodes a deictic value, such as *hi* or *saw*; this class subsumes both the definite format NP-D and the demonstrative format D-NP-D. While NP-D and D-NP-D forms built from deictic D-morphemes share certain properties, most notably rigidity effects, an NP-D form built with a deictic D-morpheme cannot be classified as a demonstrative on that basis alone. The reason is straightforward: NP-D, whether formed with a deictic or anaphoric D-morpheme, cannot be used contrastively. D-NP-D, on the other hand, whether deictic or anaphoric, cannot be used in globally unique contexts. Both of these distributional restrictions are illustrated in §2.3 and summarized in Table 1.

in Mizo. Under the analyses discussed above, we would expect pointing to contribute at-issue meaning only in the case of deictic demonstratives (D-NP-D), and not deictic definites (NP-D). However, the Mizo data show that *both* configurations render the pointing at-issue, such that an identity inference must obtain between the demonstratum and the description in both cases. Consider the example below.

(28) *Context: There is a farewell event at the school. There are two senior teachers, Teacher 1 and Teacher 2, but the farewell is for Teacher 2's retirement. The speaker and two of her friends—Friend 1 and Friend 2—are standing together in the auditorium. Friend 2 is unaware of the occasion and asks what the event is for. Friend 1, mistakenly assuming that the farewell is for Teacher 1, points at Teacher 1 and says:*

- a. *zirtirtu saw_→ a pension dawn*
 teacher DIST 3.SG pension MOD
 ‘The teacher is retiring.’
- b. Speaker: No, that's not the teacher who is retiring
- c. Speaker: Yes, but you're pointing at the wrong teacher

In this context, the speaker can correct Friend 1's mistake using (28b), but not (28c): since pointing composes noun-internally in deictic D-expressions, there is no separation between the pointing and the descriptive content available, making (28c)'s attempt to accept the description while correcting the pointing infelicitous. As expected, similar patterns hold for deictic demonstratives. Moreover, the rigidity effects identified by Kaplan (1989), exemplified in (29), are observed with both deictic definites and demonstratives in Mizo.

(29) *Context: Charles is from Charleston, West Virginia. Paul is from St. Paul, Minnesota. δ is a pointing by the speaker in the direction of Paul, who is seated on a chair in front of the speaker. The speaker says: **Look over here.** [THE GESTURE δ IS INITIATED NOW AND HELD THROUGHOUT THE NEXT SENTENCE.] **If Charles and Paul had changed chairs, then***

- a. the man being pointed at would be from Charleston.
- b. this man being pointed at would be from Charleston. (from Elbourne 2008: 440)

In this context, Kaplan rightly points out that (29a) is true and (29b) is false. In Mizo, both deictic definites and demonstratives would yield the same truth value — *false*. Thus both (30a) and (30b) would be judged false.⁶

(30) *Context: Two women—one Indian and one Nepali—are seated next to each other, with the Nepali woman sitting to the right of the Indian woman. The speaker points at the Nepali woman and says: **If they exchanged seats...***

⁶Similar patterns have been noted for Iberian Spanish in Bombi (2025) for the two different deictic constructions in the language: for example, *esto libro* DEM book' and *el libro este* DEF book DEM', although Bombi points out that prosodic focus plays an important role in distinguishing the two. It remains to be investigated how these constructions pattern relative to Mizo.

- a. *dinglam-ah hmeichhia saw_→ Indian a ni ngei ang*
 right-LOC woman DIST Indian 3.SG is MOD MOD

‘The woman on the right (who I’m pointing at) would have been Indian.’
 (judged false, since the person being pointed at is Nepali regardless of where she sits.)

- b. *saw dinglam-ah hmeichhia saw_→ Indian a ni ngei ang*
 DIST right-LOC woman DIST Indian 3.SG is MOD MOD

‘That woman on the right (who I’m pointing at) would have been Indian.’
 (judged false, since the person being pointed at is Nepali regardless of where she sits.)

The Mizo facts support Ahn’s (2025) proposal that pointing composes noun-internally, and further indicate that this effect is mediated by a linguistic exponent that parallels gestural pointing. Crosslinguistically, such exponents are typically realized in demonstrative determiners. The Mizo pattern, however, shows that they are not exclusive to demonstratives and may instead surface in other D-expressions, such as deictic definites in Mizo (and other Kuki-Chin languages). The two diverge, however, in their ability to support contrastive use: while both are rigid under pointing, only demonstratives permit contrastive interpretations. I return to this pattern in §3.3, where I argue that this contrast results from demonstratives encoding an anti-uniqueness presupposition.

Returning to anaphoric, or index, D-morphemes such as *chu* in Mizo, we observe the reverse of the deictic pattern: they are restricted to anaphoric use and cannot be employed deictically. In particular, *chu* is infelicitous in contexts involving objects in the immediate perceptual field of the interlocutors, as illustrated in (31a). This distribution follows if *chu* is an exponent of $D:USr+\mathcal{I}$.

- (31) *Context: A group of guests (who are situated at a distance from the speaker and addressee) are leaving the speaker’s place.*

- a. *#mikhual-chu an chhuak dawn*
 guest-INDEX 3.PL leave MOD

Intended: ‘The guests are leaving.’

- b. *mikhual-saw an chhuak dawn*
 guest-DIST 3.PL leave MOD

‘The guests are leaving.’ (Mizo)

In fact, the shared characteristics that English *the* and *that* exhibit—such as covariability under intensional operators, discussed by King (2001) and Elbourne (2008)—are likewise observed in Mizo DPs marked with *chu*, including both definites and demonstratives. As shown in (32a) and (33a) below, both NP-*chu* and *chu*-NP-*chu* can co-vary depending on the world or situation of evaluation. As predicted, deictic definites and demonstratives are ruled out in such contexts, as in (32b) and (33b).

- (32) Narrow scope with intensional predicates: ✓ INDEX Ds, ✗ DEIXIS Ds

- a. *kum thum hnua PM thar kan nei chu-an, PM chu/ chu PM chu*
 year three after PM new 1.PL HAVE TOP-ERG, PM INDEX/ INDEX PM INDEX
hmeichhia a ni ngei ka beisei
 woman 3.SG is MOD 1.SG hope

‘If we have a new Prime Minister in three years, I hope the Prime Minister is a woman.’

- b. *kum thum hnua PM thar kan nei chu-an, #{PM hi}/ #{he PM hi}*
 year three after PM new 1.PL HAVE TOP-ERG, PM PROX/ PROX PM PROX
hmeichhia a ni ngei ka beisei
 woman 3.SG is MOD 1.SG hope

Intended: ‘If we have a new Prime Minister in three years, I hope the Prime Minister is a woman.’
 (Mizo)

(33) Covariation: ✓ INDEX Ds, ✗ DEIXIS Ds

- a. *Mikhual a lo kal chu-an, mikhual chu/ chu mikhual chu i*
 guest 3.SG MOD come TOP-ERG, guest INDEX/ INDEX guest INDEX 2.SG
lo be dawn ania
 MOD talk MOD BE

‘If a guest comes, you should greet them.’

- b. *Mikhual a lo kal chu-an, #{mikhual saw}/ #{saw mikhual saw} i*
 guest 3.SG MOD come TOP-ERG, guest DIST/ DIST guest DIST 2.SG
lo be dawn ania
 MOD talk MOD BE

Intended: ‘If a guest comes, you should greet them.’
 (Mizo)

In the spirit of Kaplan, I argue that when a D-expression encodes only a deictic property together with an ι operator, it rigidly designates an individual, as in the case of deictic definites and deictic demonstratives in Mizo. By contrast, when a D-expression encodes an index in addition to the ι operator, it yields an anaphoric interpretation, since the index allows co-indexation with higher operators and facilitates covariability. On this view, the dual character observed in demonstratives in several languages, including English, arises because index and deixis are not realized as distinct morphemes. When both a deictic property and an index are encoded within a single D-expression, it is compatible with both deictic and anaphoric uses and can therefore be employed in either function.

In the next section, I make this distinction explicit by providing definitions of deictic and anaphoric D-expressions in Mizo. I propose that in the case of deictic D, the referent is the unique individual that bears the relevant deictic or spatial relation to the speaker, thereby situating the entity within the utterance context. Anaphoric D-expressions, by contrast, encode an index that

<i>Effects</i>	Definite		Demonstrative	
	<i>Deictic</i>	<i>Anaphoric</i>	<i>Deictic</i>	<i>Anaphoric</i>
<i>Rigidity</i>	✓	✗	✓	✗
<i>Covariability</i>	✗	✓	✗	✓

Table 2: Rigidity and covariability across definite and demonstrative D-expressions in Mizo

permits co-indexation with operators taking wide scope over them, yielding the covariability effects characteristic of anaphoric interpretations. I return to a discussion of English demonstratives in §4.1.

3.2 Deriving D-expressions in Mizo

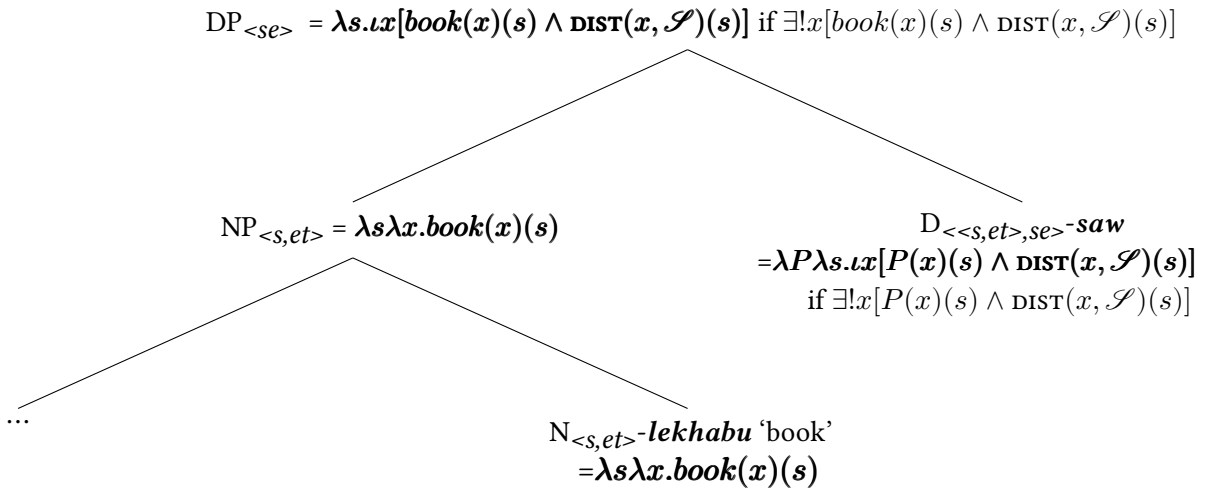
We saw that definite and demonstrative descriptions in Mizo are built from distinct morphosyntactic configurations that transparently encode deixis and index. This decomposition makes it possible to isolate the contribution of each component and, in particular, to identify what distinguishes demonstratives from definites. In the following, I build on this foundation to show that the distinctions between demonstrative and definite constructions in Mizo follow from the encoding of anti-uniqueness. Within the (de)compositional framework I have advanced, deictic definites host the D-morpheme $\mathbf{D}:U_{s_r} + \mathcal{R}$, and anaphoric definites host $\mathbf{D}:U_{s_r} + \mathcal{I}$.

(34) Definition of deictic definites in Mizo

$\llbracket \mathbf{D}_{\mathcal{R} \langle \langle s, et \rangle, se \rangle} \rrbracket^{c,g} = \lambda P \lambda s. \iota x [P(x)(s) \wedge \mathcal{R}(x, \mathcal{S})(s)]$ if $\exists! x [P(x)(s) \wedge \mathcal{R}(x, \mathcal{S})(s)]$,
undefined otherwise.

where $\mathcal{R}$ is the deictic relation encoded in D and $\mathcal{S}$ is the speaker

(35) Deriving *lekhabu saw* ‘book D_{DIST} ’



‘The unique individual that is a book situated at a location far from the speaker in the utterance context s , defined if there is exactly one book that is situated at a location considerably far from the speaker in the utterance context s ’

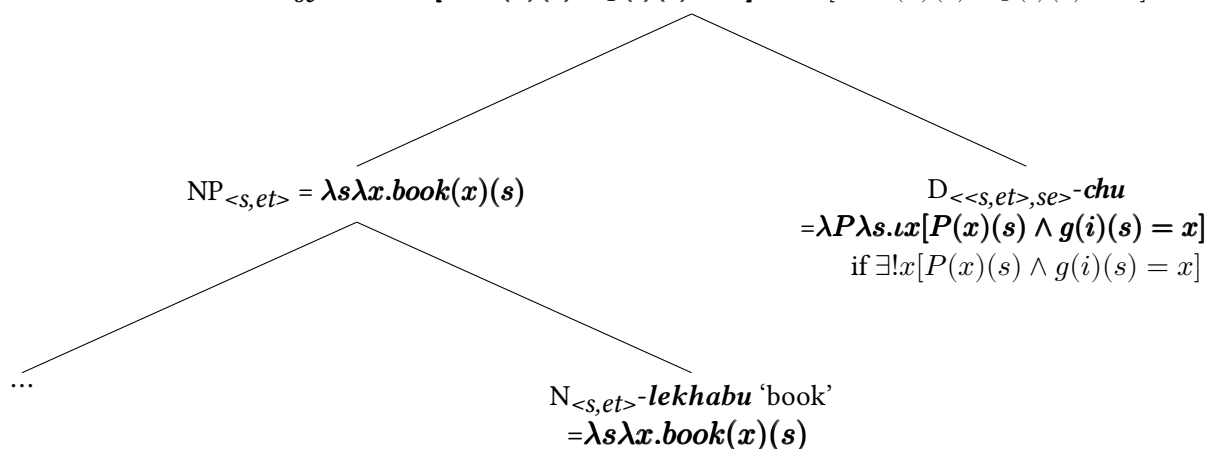
(36) **Definition of anaphoric definites in Mizo**

$$\llbracket D_{i < \langle s, et \rangle, se} \rrbracket^{c, g} = \lambda P \lambda s. \iota x [P(x)(s) \wedge g(i)(s) = x] \quad \text{if } \exists! x [P(x)(s) \wedge g(i)(s) = x],$$

undefined otherwise.

where i is an index

(37) Deriving *lekhabu chu* ‘book D_i ’

$$\text{DP}_{\langle se \rangle} = \lambda s. \iota x [\text{book}(x)(s) \wedge g(i)(s) = x] \text{ if } \exists! x [\text{book}(x)(s) \wedge g(i)(s) = x]$$


‘The unique individual that is a book that is assigned to the index i in the utterance context s , defined if there is exactly one book that is assigned to the index i in the utterance context s .’

I now turn to demonstrative constructions in Mizo, which are expressed in the D-NP-D configuration. A question arises: why are the two Ds in D-NP-D constructions always the same? They strictly agree in their deictic value and mismatched combinations, as in (38), are ungrammatical.

- (38) **he* *lekhabu saw/* **saw* *lekhabu hi/* ...
 MOD:PROX book D_{DEM}:DIST/ D_{DEM}:DIST book MOD:PROX

Moreover, the two D-morphemes differ tonally: the pre-nominal carries a rising tone while the post-nominal carries a flat tone.⁷ I claim that the pre-nominal D is a reduplicative morpheme, RED, with a floating tone, drawing its PF features from the post-nominal D-morpheme, in the

⁷While Mizo is a tonal language, the extent to which tone is lexical remains unclear at this stage of my fieldwork. This does not, however, affect the analysis, since the two D-morphemes are semantically distinct: the pre-nominal is an NP-modifier carrying an anti-uniqueness presupposition, while the post-nominal is a D-morpheme encoding the ι operator together with either a deictic value or an anaphoric index. The key point is that the pre-nominal and post-nominal D share the same PF features, differing only in tone.

spirit of Paster’s (2003) analysis of Gã. It functions as an identity function that enriches the NP with an anti-uniqueness presupposition, with the tonal difference attributable to its floating tone. This is supported by the fact that the tone of RED differs depending on whether an NP intervenes: it is rising in D-NP-D (*he rose hi* ‘this rose’) and falling in D-D (*he hi* ‘this’). The reduplicative status of the pre-nominal morpheme is further corroborated by the illicit status of bare D-NP constructions (**he lekhabu* ‘prox-book’). Within the (de)compositional framework then, RED in Mizo simply captures one of the four definiteness semantic blocks: $D:AU_{s_l}$. The post-nominal D then turns the complex NP into a referential expression, adding either anaphoric information or a deictic property to it. The derivations of deictic and anaphoric definites and demonstratives are illustrated in (40) and (43) below.

(39) Definition of NP-modifier RED in Mizo

$$\llbracket \text{RED}_{\langle\langle s, et \rangle, \langle s, et \rangle \rangle} \rrbracket^{c,g} = \lambda P \lambda x. P(x) \text{ if } \exists s' s \leq s' \mid |P(x)(s')| > 1, \\ \text{where } s \text{ is provided by the utterance context}$$

(40) Deriving deictic demonstrative *saw lekhabu saw* ‘RED book D_{DIST} ’

$$\begin{aligned} \text{DP}_{\langle se \rangle} &= \lambda s. \iota x [\text{book}(x)(s) \wedge \text{DIST}(x, \mathcal{S})(s)] \text{ if } \exists! x [\text{book}(x)(s) \wedge \text{DIST}(x, \mathcal{S})(s)] \\ &\quad \wedge \exists s' s_r \leq s' \mid |\text{book}(x)(s')| > 1 \\ &\quad \swarrow \quad \searrow \\ \text{NP}_{\langle s, et \rangle} &= \lambda s \lambda x. \text{book}(x)(s) \quad \quad \quad \text{D}_{\langle\langle s, et \rangle, se \rangle} \text{-saw} \\ &\text{if } \exists s' s_r \leq s' \mid |\text{book}(x)(s')| > 1 \quad \quad \quad = \lambda P \lambda s. \iota x [P(x)(s) \wedge \text{DIST}(x, \mathcal{S})(s)] \\ &\quad \quad \quad \text{if } \exists! x [P(x)(s) \wedge \text{DIST}(x, \mathcal{S})(s)] \\ &\quad \swarrow \quad \searrow \\ \text{RED}_{\langle\langle s, et \rangle, \langle s, et \rangle \rangle} &\quad \quad \quad \text{N}'_{\langle s, et \rangle} \text{-lekhabu ‘book’} \\ &= \lambda P \lambda s \lambda x. P(x)(s) \quad \quad \quad = \lambda s \lambda x. \text{book}(x)(s) \\ &\text{if } \exists s' s \leq s' \mid |P(x)(s')| > 1 \end{aligned}$$

PF: RED $\leftrightarrow$ *saw* (rising tone)

‘The unique individual that is a book situated at a location considerably far from the speaker in the utterance context s , defined if s is an alienable minimal situation⁸ in which there is exactly one book situated in the distal location and⁹ a situation larger than s has at least two books.’

⁹Following Karttunen and Peters (1979) and Beaver and Krahmer (2001), I treat presuppositional content as orthogonal to the function application that computes at-issue content. Function application operates solely on the at-issue semantic contribution of the expressions involved, while presuppositions project independently and accumulate as a conjunction. As shown in (40), the presupposition of the derived DP is simply the conjunction of the uniqueness presupposition contributed by the definite determiner *saw* ‘ D_{DIST} ’ and the anti-uniqueness presupposition contributed by the NP-modifier represented by the reduplicative morpheme RED. Both are conjoined at the DP node without either interfering with the computation of the at-issue extension. See Heim (1983) for the dynamic semantics implementation of the same insight

I argue that the sequence RED–NP in Mizo is not licensed as an argument on its own, since it is not referential. To acquire referential force, it must be bound by an ι operator. However, covert ι type-shifting is unavailable in Mizo, because the presence of overt D morphemes blocks this operation, in accordance with Chierchia’s (1998) *Blocking Principle*, given in (41) below.

(41) *Blocking Principle* (‘Type-shifting as Last Resort’)

For any type-shifting operator τ and any X : ${}^*\tau(X)$, if there is a determiner D such that for any set X in its domain, $D(X) = \tau(x)$. (Chierchia 1998:360)

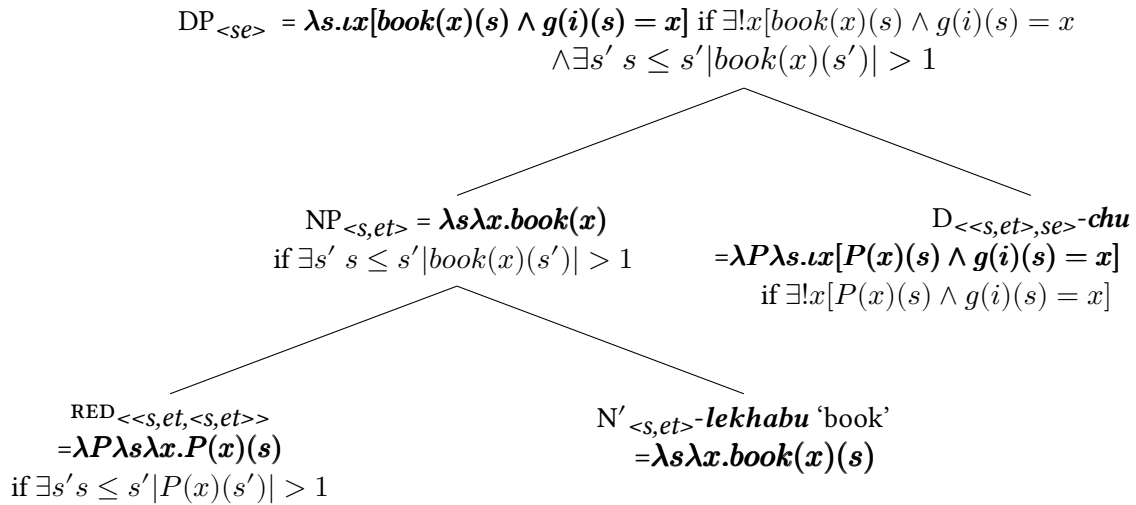
Mizo demonstrative morphemes are reduplicative, and in the absence of the post-nominal definite determiner, it is not morphologically transparent whether a Dem–NP sequence is independently available. Nevertheless, the present analysis predicts that such sequences should not function as referential descriptions in languages like Mizo. This prediction is borne out in Hakha Chin, another Kuki-Chin language that employs deictic definites. Hakha Chin has an overt demonstrative morpheme, *mah*, and, as predicted, a Dem–NP sequence cannot function as a referential expression in the absence of the post-nominal definite determiner, as in (42) below.

- (42) a. ***mah*** *uico* **(hi)* *a-lian*
 DEM dog PROX 3.SG-be.big

Intended: ‘This dog is big.’

(*Hakha Chin*; adapted, Wamsley 2019:6)

(43) Deriving anaphoric demonstrative ***chu lekhabu chu*** ‘RED book D_i ’



PF: RED $\leftrightarrow$ ***chu*** (rising tone)

‘The unique individual that is a book that is assigned to the index i in the topic situation s , defined if s is an alienable minimal situation in which there is exactly one book that is assigned to the index i , and a situation larger than s has at least two books.’

3.3 The role of anti-uniqueness

In §3.1, I put forward a set of arguments which, taken together, show that the division between definite and demonstrative expressions cannot be reduced to deixis or index. In particular, we saw that not only demonstratives, but also deictic definites, exhibit name-like rigidity and, in the Kaplanian sense, function as rigid designators. The distinctions typically observed between *the* and *this* in English with respect to rigid reference or at-issue pointing (Ebert, Ebert and Hörnig 2020; Ahn 2022) do not hold in Mizo. Both deictic definites and deictic demonstrative constructions display the properties associated with the deictic use of English *that*. Where they instead come apart is in their contrastive potential and in their compatibility with globally or functionally unique descriptions. D-NP-D constructions can be used contrastively in contexts where more than one entity in the utterance context satisfies the relevant property, whereas NP-D constructions cannot. Conversely, D-NP-D constructions are infelicitous with globally or functionally unique descriptions, while NP-D constructions are not. In what follows, I show how the anti-uniqueness presupposition derives these distinctions.

3.3.1 Global/functional uniqueness

The distinction at issue is the following: constructions such as *saw-ni-saw* ‘RED-sun-DIST’ are infelicitous in ordinary contexts,¹⁰ as illustrated in (44b) below. By contrast, *ni-saw* ‘sun-DIST’ is not subject to the same restriction (44c).

- (44) a. The/ #That sun is shining bright.
 b. #*{saw ni saw}* *en teh*
 RED sun DIST look IMP
 Intended: ‘Look at the sun (over there)!’
 c. *ni saw en teh*
 sun DIST look IMP
 ‘Look at the sun (over there)!’ (Mizo)

Similarly, an expression such as *chu-referee-chu* ‘RED-referee-INDEX’, when used in connection with a specific football match, is infelicitous (45b), whereas *referee-chu* ‘referee-INDEX’ is not, as shown in (45c) below.

- (45) Context: *The speaker is reporting something to the addressee that she heard from her friend Dee.*
She says: Dee was watching a football match,...
 a. The/ #That referee was partial.
 b. #*{chu referee chu-an}* *tran lam a nei*
 INDEX referee INDEX-ERG support side 3.SG have
 Intended: ‘The referee was partial.’

¹⁰The same exceptions discussed for English demonstratives also arise in Mizo. For instance, *chu-ni-chu* ‘RED-sun-INDEX’ is felicitous in a context in which an astronomer is making observations about another solar system. A second exception involves the emotive use.

- c. *referee chu-an tran lam a nei*
 referee INDEX-ERG support side 3.SG have

‘The referee was partial.’ (Mizo)

The effect in these cases is straightforward: the anti-uniqueness presupposition introduced by the pre-nominal modifier RED rules out the felicitous use of D–NP–D constructions. The definition of RED in Mizo is repeated in (46) below.

- (46) $\llbracket \text{RED} \rrbracket^{c,g} = \lambda P \lambda x. P(x)$ if $\exists s' s_r \leq s' \mid x : P(x)(s') \mid > 1$,
 where s_r is provided by the utterance context

For (44), given that the largest situation by default is one that is within our solar system, the description *ni sun*’ is globally unique, and thus *saw-ni-saw RED-sun-DIST*’ is not felicitous. The same reasoning extends to functionally unique predicates such as *referee* (of a football match). The reduplicative pre-nominal modifier (RED) in *chu-referee-chu* ‘RED-referee-INDEX’ introduces an anti-uniqueness presupposition requiring the existence of at least two referees in some larger situation. This presupposition cannot be satisfied, since there is no larger situation in which the antecedent football match has more than one referee.

3.3.2 Contrastive use

As noted above, a key distinguishing feature between demonstratives and definites is the former’s ability to be used contrastively (highlighted by Dayal and Jiang 2022), illustrated again in (47). Crucially, definites do not exhibit this property (47b).

- (47) a. This dog is sleeping and this dog is running around.
 b. #The dog is sleeping and the dog is running around

Dayal and Jiang (2022) attribute the availability of contrastive uses with demonstratives to the assignment of distinct indices to the intended referents. However, as shown in §3.1, deictic definites and demonstratives in Mizo do not involve anaphoric use, and as a result do not introduce indices. Nevertheless, demonstratives can still be used contrastively, as in (49a) below, whereas deictic definites crucially cannot (48b).

- (48) a. *he ui hi_{→A} a mu leh he ui hi_{→B} a tlan*
 RED dog PROX 3.SG sleep and RED dog PROX 3.SG run

‘This dog_{→A} is sleeping and this dog_{→B} is running.’

- b. *#ui hi_{→A} a mu leh ui hi_{→B} a tlan*
 dog PROX 3.SG sleep and dog PROX 3.SG run

Intended: ‘This dog_{→A} is sleeping and this dog_{→B} is running.’ (Mizo)

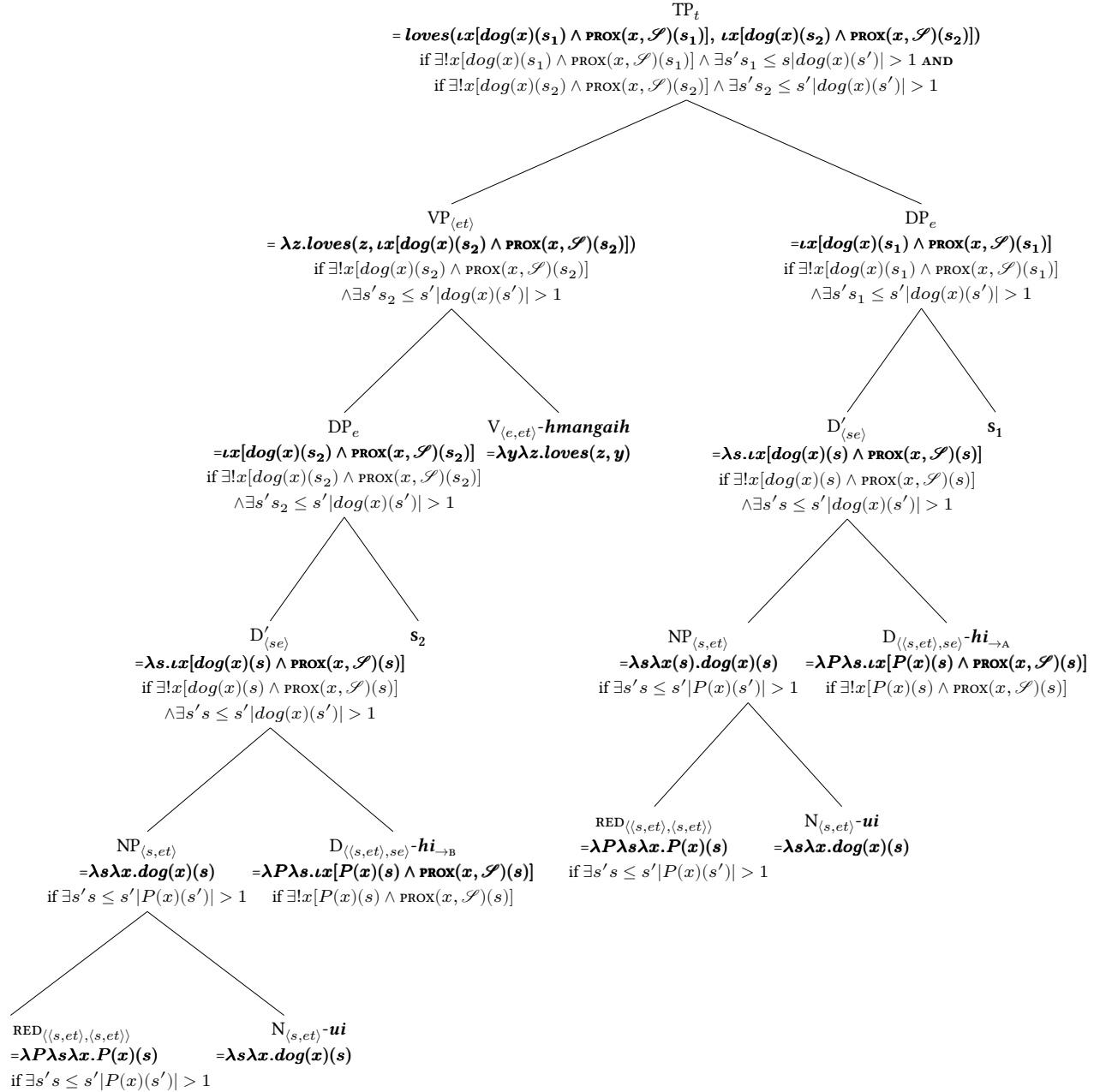
In the absence of index, how would this be possible? I propose that AU and situational restriction alone suffice to derive the desired result. Consider the example in (49) below.

- (49) a. *he ui hian_{→A} he ui hi_{→B} a hmangaih*
 RED dog PROX-ERG PROX dog PROX 3.SG love
 ‘This dog_{→A} loves this dog_{→B}.’

I suggest that the utterance context provides two restricted situations corresponding to the two demonstrative descriptions — two alienable minimal situations, each containing a unique pointing associated with a unique dog. This situation-splitting is licensed precisely by the richer presuppositional profile of D-NP-D. Recall that D-NP-D carries two presuppositions: the uniqueness presupposition contributed by the post-nominal D, and the anti-uniqueness presupposition contributed by RED. It is the latter that does the crucial work here: by requiring the existence of a larger situation in which the NP property is satisfied by more than one individual, the anti-uniqueness presupposition forces the utterance context to be partitioned into sub-situations, each containing a distinct demonstratum. NP-D, which lacks this presupposition, evaluates uniqueness against a single restricted situation and therefore cannot license such a split — which is why it fails in contrastive contexts. These restricted situations are sub-situations to which the speaker directs the interlocutors’ attention in the individual demonstrations — this $X \rightarrow A$ (sub-situation pointing at dog A) and this $X \rightarrow B$ (sub-situation pointing at dog B). In each case, the identity of the demonstratum is resolved locally within the relevant sub-situation: it is the unique individual that bears the relevant NP and deictic property and is identical to the entity being pointed at (drawing on Ebert, Ebert and Hörnig 2020, Ahn 2022; Ahn 2025). In deictic uses of the kind illustrated in (49), the sub-situation variable saturates the DP prior to its composition with the verb. The anti-uniqueness presupposition is satisfied because, in the larger situation — namely, the one in which both demonstrations occur — there are two entities that satisfy the NP property *dog*. The derivation of (49) is illustrated below in (50). Note that in (50), the situations s_1 and s_2 are *sub-situations*, and the entire utterance context is the larger situation s' . The rigidity of deictic definite descriptions follows from the fact that the sub-situation against which uniqueness is evaluated is fixed by a pointing act in the utterance context. Since this sub-situation is strictly anchored to the utterance context, it cannot be bound higher up by intensional operators. The deictic definite description, therefore, rigidly denotes the unique entity that is the demonstratum within the sub-situation established by the pointing act.

For presentational clarity, I have shown the composition at the base positions of the subject and object in Mizo; composition at the moved positions would instead involve Predicate Abstraction at the Topic Phrase projections. Additionally, Vocabulary Insertion for RED is determined by the lexical entry of the corresponding D head, yielding *he* ‘PROX’ with a rising tone in both positions in the present case.

- (50) *translates to:* A unique dog located proximally to the speaker in the sub-situation s_1 loves a unique dog located proximally to the speaker in the sub-situation s_2 , defined *if* there is a unique dog proximal to the speaker in the sub-situations s_1 **and** s_2 , and there is a situation s' larger than both s_1 **and** s_2 that has more than 1 dog.



In fact, the contrastive use pivots on both the AU presupposition and co-speech pointing. In the absence of the former, contrast is not possible in Mizo, as in (47b). Similarly, in the absence of the latter, contrastive use is not felicitous. This is not specific to Mizo, but holds for English as well, as shown in (51) below.

- (51) a. Yesterday I saw a dog_i getting introduced to another dog_{i'}. ???That dog_i jumped on that dog_{i'}.
- b. If a bishop_i meets another bishop_{i'}, ???that bishop_i greets that bishop_{i'}.

Therefore, the contrastive use of demonstratives pivots on two conditions: the anti-uniqueness presupposition and co-speech gesturing.

3.4 Zooming out

The main contribution of the analysis is that it provides a way of understanding the individual pieces that constitute demonstratives, and D-expressions more generally. I have argued that four independent components are at play, and that different combinations of these yield different D-expressions: uniqueness in the restricted situation (U_{S_r}), an anaphoric index ($\mathcal{I}$), a deictic property ($\mathcal{R}$), and anti-uniqueness in a larger situation (AU_{S_l}). While this compositional structure is not morphologically transparent in English, some languages — such as Mizo and other Kuki-Chin languages — make these components overt.

The analysis extends Dayal and Jiang’s (2022) insight that the definite–demonstrative distinction tracks two environments — contrastive uses and uses with globally or functionally unique descriptions — by grounding it in the presence or absence of an anti-uniqueness component. Specifically, I argue that AU_{S_l} is the component responsible for this pattern: it accounts for why globally or functionally unique descriptions are infelicitous with both anaphoric and deictic demonstratives, while remaining fully acceptable with definites. At the same time, in deictic demonstratives, AU_{S_l} underlies the availability of contrastive uses — a property that definites lack.

Several implications follow. First, AU_{S_l} is necessary for a genuine demonstrative use; deixis alone cannot suffice. The observed distinction between demonstratives and definites therefore cannot be reduced to a pragmatic effect of pointing, as proposed in Modality-Linking theories. At the same time, Modality-Linking theories rightly emphasize the role of co-speech pointing in contrastive uses: pointing makes an at-issue contribution and fixes the identity of the referent. However, co-speech pointing alone cannot derive contrast. As I have shown for Mizo, deictic definites cannot be used contrastively, even though they involve at-issue pointing. The presence of pointing alone is therefore insufficient to derive contrastive use. What distinguishes demonstratives is their ability to contrast, and this contrastive capacity is a direct semantic consequence of AU_{S_l} . A further implication concerns the decoupling of deictic and anaphoric uses of demonstratives. Deictic uses exhibit rigidity, in agreement with Kaplan’s classical view, whereas anaphoric uses give rise to the covariability observed with demonstratives such as *that* in English. The contrast between rigidity and covariability thus tracks the presence or absence of index, rather than demonstrative status per se. Crosslinguistically, languages vary in how D-expressions are realized. With respect to definite descriptions, some languages employ overt determiners, while others rely on ι type-shifting with bare nouns (e.g., Mandarin, Hindi, Russian). In the article-less languages, bare nouns correspond to both unique and anaphoric interpretations, although the latter may also be expressed with demonstratives (Jenks 2018, Dayal and Jiang 2022). Across both language types, demonstrative expressions are typically realized by a dedicated morphological form. While the typological tendency is to realize both anti-uniqueness and ι in a single exponent, such as English *this* or *that*, certain languages, such as Mizo and Akan (Dayal *in prep*; Owusu 2022; Owusu 2025), deploy separate morphemes: one contributing ι , and another introducing an anti-uniqueness presupposition — realized as a DP modifier in Akan (Owusu 2025) and as an NP modifier in Mizo, as illustrated in §34. In the next section, I discuss some of these languages with transparently compositional demonstratives.

4 D-expressions typologically

Although deictic definites, as found in Mizo and other Kuki-Chin languages (Wamsley 2019), appear to be typologically rare, a number of languages nevertheless instantiate distinct morphological exponents corresponding to different building blocks of D-expressions. The way languages package the four semantic blocks to form different d-expressions involves a range of crosslinguistic variation. In what follows, I present preliminary data from a small set of languages to briefly examine how the proposal developed in the previous section extends to these systems. I begin with English as a baseline, focusing on demonstratives that are not transparent, in which a single morphological exponent that expresses all four D-expression properties. I then turn to three languages: Greek, Akan, and Bangla, that realize these properties through distinct D-morphemes, each encoding different combinations of the relevant features.

4.1 English: fused demonstratives

Demonstratives in English combine all four D-expression properties into a single morphological exponent. Presumably, this lies at the core of debates about the different uses of demonstratives, particularly the contrast between Kaplan-style deictic uses and referential non-deictic uses - which one is quintessentially a ‘demonstrative’? Crucially, as Robinson (2005) and Dayal and Jiang (2022) observe, deixis alone cannot explain the disparity in cases like (52). English demonstratives encode an anti-uniqueness condition, which restricts their distribution in cases where the referent, qua NP property, is strictly unique across all relevant larger situations. In such contexts, demonstratives are infelicitous but the definite determiner is not (53).

- (52) a. #That Finance Minister proposed a terrible budget. b. #That sun rises in the east.
- (53) a. The Finance Minister proposed a terrible budget. b. The sun rises in the east.

With these considerations in place, I briefly revisit the distinction between *this* and *that* in English, extending the Mizo-based decomposition to these two forms. The discussion here is necessarily tentative: unlike Mizo, English does not morphologically separate deictic and anaphoric D-morphemes, so the decomposition must be inferred from distributional evidence rather than read off directly from the morphology. Elbourne (2008) observes that *this* is markedly degraded, relative to *that*, when it occurs outside the scope of a propositional attitude predicate, as illustrated in (54).

- (54) a. ??Mary talked to no senator without *this* senator thinking she was crazy.
b. Mary talked to no senator without *that* senator thinking she was crazy.

One possible interpretation of this contrast, consistent with the present framework, is that *this* does not encode an anaphoric index, patterning in this respect with Mizo deictic D-morphemes, while *that* encodes both a deictic property and an index. On this view, the degraded status of (54a) would follow from the absence of an index in *this*, which prevents it from co-varying with the universal quantifier. When the referent of the demonstratum is fixed by a pointing act in the utterance context, as in (55), the degradation disappears, since rigid designation suffices for reference.

- (55) Mary talked to no senator without *this* (pointing at *A*, who is in the utterance context) senator thinking she was crazy.

In Elbourne’s analysis, a demonstrative such as *this* or *that* encodes an ι operator with three restrictions: an NP property, a relational component (deixis), and an index, as given in (56) below.

- (56) a. [DP [[*that* i] R] NP]
 b. $\llbracket \textit{that} \rrbracket^{w,a,t,h} = \lambda x. \lambda f. \lambda g. \lambda s. \iota z [f(x)(\lambda s'. z)(s) = 1 \wedge g(\lambda s'. z)(s) = 1 \wedge \textit{distal}(x, w, a, t)]$,
 where w is the possible world where the utterance takes place, a is the agent, t is the time of utterance, and h is the variable assignment.

In light of data such as (54), a possible lexical entry for *this* parallels that of deictic demonstratives in Mizo in that it lacks a separate argument for index and also encodes an identity inference. Further nuance emerges from certain distributional properties of *that*. Ahn and Davidson’s (2018) experimental work examines the interaction between pointing and the availability of anaphoric readings of *that*. They show that in the presence of co-speech pointing, the deictic use of *that* is privileged over a potential anaphoric interpretation, as illustrated in (57a) below. Ahn (2022) provides more examples of this phenomenon, one of them is illustrated in (57b).

- (57) a. Every time I met a girl, I talked to that girl_→ (Ahn and Davidson 2018)
 b. One woman is my friend. That woman_→ / That woman plays soccer. (Ahn 2022: 1365)

One possibility, then, is that *that* is ambiguous between a deictic and an anaphoric interpretation. On this view, English would largely parallel the pattern observed in Mizo, despite the morphologically fused nature of its D-expressions. Another possibility is that the presence of co-speech pointing situates the relational property (R) within the utterance context. Following Elbourne’s formalism, in cases such as (57b), *That woman*_→ denotes the unique individual in the utterance context that satisfies the NP property *woman* and also satisfies, in the utterance context, the property obtained by composing the relational component R (a DISTAL property) with an index i . Even if *That woman*_→ were to inherit the index from the preceding discourse, it would fail to refer to the previously mentioned *One woman* once its relational property is fixed in the utterance context through co-speech pointing, making such an use infelicitous. What, then, about purely anaphoric uses of demonstratives? Elbourne (2008) suggests that the relational property R in such cases is not a proximal or distal one, but something much more trivial. For example, he suggests that for a demonstrative construction like (58) below, R could “take on as its value the function mapping utterances to the property of having one’s existence announced in them” (Elbourne 2008: 439).

- (58) $\llbracket \llbracket \textit{that } i_1 \rrbracket R_2 \rrbracket$ [student who scored 100 on the exam] (Elbourne 2008: eg 114, pg 439)

Extending that to cases like (57b), we could argue that in the evaluation of the anaphoric case *That woman* (without pointing), R is an arbitrary property. On the other hand, R in *That woman*_→ is the property of being the distal demonstratum at the utterance context. Thus the identity of the referent in the anaphoric use *That woman* and deictic use *That woman*_→ does not have to be the same, and in fact it would be odd if it is. We can avoid positing an ambiguity analysis for *that*

once we consider that its core semantics does not determine the value of R , beyond specifying that it is a relational property of some kind. In the absence of co-speech gesturing, this property is trivially satisfied, as proposed by Elbourne (2008). In the presence of co-speech gesturing, however, R is necessarily resolved as the deictic property DIST. Within the (de)-compositional framework advanced in this paper, I posit the following entry for English *Dem*.

- (59) $\llbracket \text{Dem} \rrbracket^{c,g} = \lambda P \lambda s. \iota x [P(x)(s) \wedge \mathcal{R}(x, \mathcal{S})(s) \wedge g(i) = x]$
 if $\exists! x [P(x)(s_r) \wedge \mathcal{R}(x, \mathcal{S})(s_r) \wedge g(i) = x] \wedge \exists s' s \leq s' |P(x)(s')| > 1$, undefined
 otherwise.

where $\mathcal{R}$ = relational property, g = assignment function, c = evaluation context, i = index, and $\mathcal{S}$ = speaker

It is plausible that the distinction between *this* and *that* follows from how $\mathcal{R}$ is specified in a given utterance context. Crucially, when *Dem* is pronounced alongside co-speech pointing, the index is no longer assigned to the previously mentioned salient entity; instead, it must refer to the entity that is the target of the pointing or demonstration. The demonstrative *this* is the spell-out of *Dem* when it is accompanied by pointing and $\mathcal{R}$ is specified as PROX, whereas the default spell-out is *that*. I propose the following Vocabulary Insertion rules to capture this distribution.

(60) Vocabulary Insertion rules for *Dem* in English

- a. $\text{Dem}[_, \mathcal{R} : \text{proximal}] \leftrightarrow \text{this}$ b. $\text{Dem} \leftrightarrow \text{that}$

The fact that arguments of a demonstrative can be vacuously satisfied is not an implausible claim: demonstratives, in fact, can be used felicitously even in the absence of overt NP content, as in (61). Definite determiners do not share this privilege.

- (61) a. $\text{This}_{\rightarrow}$ is a cat. b. $^*\text{The}_{\rightarrow}$ is a cat.

That is, for a demonstrative to function felicitously as a referential expression, not all of its arguments must be meaningfully saturated. Importantly, the pattern is *not* arbitrary: in anaphoric uses, a demonstrative requires *both* NP content and index; in deictic uses supplemented by pointing in the utterance context, the demonstrative may suffice on its own, since the pointing establishes the identity inference that yields the kind of ‘direct reference’ described by Kaplan. In such cases, demonstratives parallel pronouns, as in (62).

- (62) a. $\text{That}_{\rightarrow}$ is a cat. b. $\text{She}_{\rightarrow}$ is a cat.

The two diverge, however, in anaphoric contexts. While both pronouns and demonstratives are saturated with an index value, a pronoun may suffice on its own, whereas a demonstrative requires its NP content, as in (63) below.

- (63) a. I made a new friend_{*i*} yesterday. $\# \text{That}_{\rightarrow}$ is a cat.
 b. I made a new friend_{*i*} yesterday. $\text{She}_{\rightarrow}$ is a cat.

However, I do not wish to overstate the case, and at present this remains speculative. Moreover, the use of *this* to refer back to a previously mentioned discourse referent is not entirely

excluded, particularly when the referent is highly salient or has been introduced very recently. In addition, this has been argued to admit cataphoric uses (Roberts 2002; Himmelmann 2011), as in (64).

- (64) a. This is an ugly word: *hippopotamus*.
 b. Do you know these new rose hybrids, Meidiland and Peace? (Roberts 2002: 122)

My aim here has been to tentatively explore the distinction between *this* and *that* by drawing a parallel with the properties exhibited by Mizo deictic and anaphoric demonstratives. A fuller investigation of English demonstratives within this framework is left for future work. In the following, I turn to two languages that more transparently instantiate a decomposed demonstrative structure.

4.2 Akan: transparent demonstratives

Akan has two determiner morphemes, *nó* and *saa*, which combine with an NP to form demonstratives, as in (65a). Similar to Mizo, Akan determiner *saa* cannot occur on its own and must be supplemented by *nó*. By contrast, *nó* can appear independently, as illustrated in (65b) and (65c) below.

- (65) a. *saa abofra #(nó) nim adee paa*
 DEM child DEF know thing INT
 ‘That child is very intelligent.’
 b. *Maame nó kasa-e*
 woman.PL DEF talk-PST
 ‘The woman spoke.’
 c. *Ama hu-u ɔkyerekyereni bi ne sogyani bi. ɔ-kyea-a sogyani*
 Ama see-PST teacher INDEF CONJ soldier. INDEF 3.SG.SUBJ-greet-PST soldier
nó
 DEF
 ‘Ama saw a teacher and a soldier. He greeted the soldier.’ (Akan, Owusu 2025)

Owusu (2025) shows that the distribution of *nó* is restricted to NPs that are not globally unique; accordingly, it displays the properties of an anti-uniqueness presupposition, as illustrated in (66). Globally unique objects, such as the sun, are instead expressed with the bare noun, which undergoes covert type-shifting.

- (66) Context: A Ghanaian newscaster is talking about the Ghanaian president.. . .
ɔmanpanin #(nó) be-bleme obi
 president DEF FUT-blame someone
 ‘The president will blame someone’ (Owusu 2025)

Building on this observation, Owusu (2025) proposes that the determiner *nó* in Akan encodes the iota operator, that carries both an anti-uniqueness presupposition and an index, as defined in (67). The anti-uniqueness component accounts for data such as (66), while the index is responsible for the availability of anaphoric readings, as in (65c).

$$(67) \quad \llbracket \mathbf{nó} \rrbracket^g = \lambda s \lambda P \lambda y : \exists! x [P(x)(s) \wedge x = y] \wedge \exists s' [s \leq s' \wedge |\{x | P(x)(s')\}| > 1].$$

$$\iota x [P(x)(s) \wedge x = y] \quad (\text{Owusu 2025})$$

Presupposes that *x* is familiar and that the cardinality of *P* in an extended situation (*s'*) is greater than 1.

By contrast, *saa* is analyzed as a DP modifier whose role is to ensure that the situation in which the uniqueness condition associated with *no* is satisfied is a sub-situation of the topic situation (). By restricting evaluation to a sub-situation of the topic situation, *saa* gives rise to the contrastive interpretation of NP-*no*. The fact that *saa* cannot occur with an NP on its own further supports its analysis as a DP modifier.

$$(68) \quad \llbracket \mathbf{saa} \rrbracket^g = \lambda W_{\langle s, \langle e, e \rangle \rangle} \lambda s'' \lambda z : \iota y [\exists s' s' \leq s'' [y = W(s')(z)]] \quad (\text{Owusu 2025})$$

I would like to propose an alternative way of capturing the Akan facts, particularly in light of the data reported by Bombi and Mac-Arthur (2025). They present fieldwork data showing that Akan permits only a restricted class of nouns to appear in bare form as referring expressions - specifically, nouns that encode a functional relation to the interlocutors, such as *president* (of the interlocutors' country), *headmaster* (of the interlocutors' school), and *teacher* (of the interlocutors' class). Moreover, bare NPs in Akan exhibit name-like scopal properties, such as rigidity across possible worlds, which makes its use in (69) infelicitous. In such cases, NP-*nó* is the choice of referring expression.

$$(69) \quad \begin{array}{llllllll} \text{Mpremprem} & \text{mu} & \text{ɔmanpanyin} & \#(\mathbf{nó}) & \text{yɛ} & \text{obarima.} & \text{Bre} & \text{bi} & \text{bɛ-ba} & \text{ɔmanpanyin} \\ \text{right-now} & \text{in} & \text{president} & \text{DEF} & \text{cop} & \text{man.} & \text{time} & \text{INF} & \text{FUT-come} & \text{president} \\ & & \#(\mathbf{nó}) & \text{be-yɛ} & \text{obaa} & & & & & \\ & \text{DEF} & \text{FUT-COP} & \text{woman} & & & & & & \end{array}$$

'Right now, the president is a man. A time will come when the president is a woman.'

(Akan, Bombi and Mac-Arthur 2025)

On the basis of this distribution, they argue for a quasi-name analysis of Akan bare nouns. Under this view, bare Ns compete with the NP-*nó* form relative to the utterance context. Following Yip et al. (2023), they further argue that it is not an anti-uniqueness presupposition that makes NP-*nó* infelicitous with globally unique objects, but pragmatic competition between the null quasi-name operator and the determiner *nó*. Specifically, cases such as (66) involve contexts in which both a uniqueness presupposition and a relation presupposition are satisfied (e.g. the unique president of the speaker's interlocutor's country). In such contexts, the choice of the bare noun follows from the principle of *Maximize Presupposition!* (Heim 1991), which states that when two forms contribute the same assertive content relative to the conversational context, the form with the stronger presuppositional requirements blocks the use of the weaker one.

In this light, we can analyze the determiner *nó* as a D-morpheme that encodes an iota operator and an index, as in (70a). The determiner *saa*, by contrast, encodes anti-uniqueness and functions as an NP modifier, as in (70b).

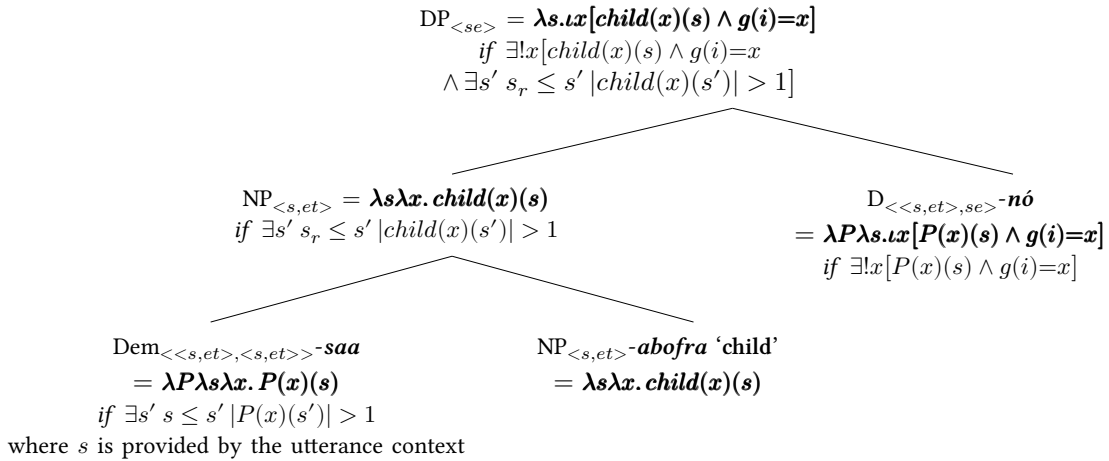
(70) Alternative analysis of Akan *nó* and *saa*

a. $\llbracket \textit{nó} \rrbracket^{c,g} = \lambda P \lambda s. \iota x [P(x)(s) \wedge g(i) = x]$ if $\exists! x [P(x)(s) \wedge g(i) = x]$, undefined otherwise

b. $\llbracket \textit{saa} \rrbracket^{c,g} = \lambda P \lambda s \lambda x. P(x)(s) \exists s' s_r \leq s' [P(x)(s') > 1]$, undefined otherwise

where s_r is the restricted situation, provided by c , in which uniqueness is evaluated. On its own, *saa* does not yield a referring expression. Given the presence of an overt D in Akan, covert type-shifting is blocked, following Chierchia's (1998) *Blocking Principle*. This is parallel to the blocking of Dem-NP in Mizo (see discussion in §3.2). Consequently, for *saa* to become a referring expression, it must combine with the determiner *nó*, yielding a structure of the form in (71) below.

(71) Deriving *saa abofra nó* 'Dem child Det'



The unique child in situation s, defined if there is exactly one child in s and there is more than one child in a situation larger than s

Under this analysis, the demonstrative systems of Mizo and Akan are formally parallel: in both languages, anti-uniqueness is contributed by a modifier that is not itself referential, while reference is achieved via combination with a determiner that introduces iota and an index. That is, in Akan, Dem hosts $D:AU_{s_l}$ and D^{11} hosts $D:U_{s_r} + \mathcal{I}$.

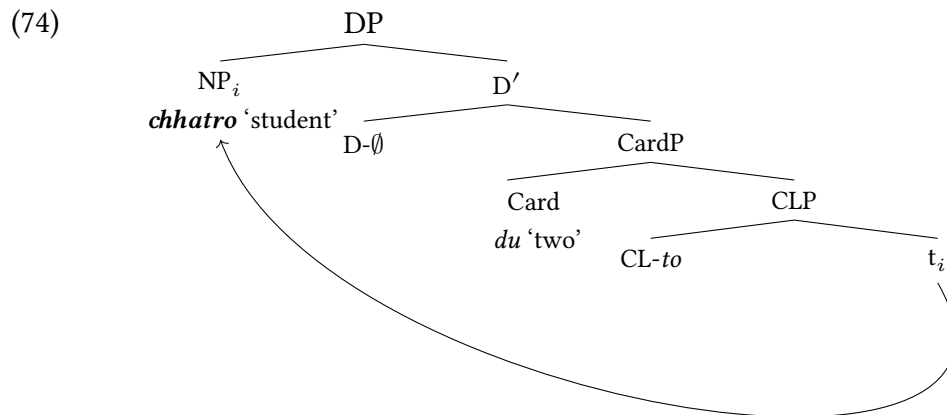
4.3 Bangla: transparent demonstratives

Like Mizo, Bangla is a classifier language. Definite descriptions in Bangla are realized as bare NP-CL constructions, as illustrated in (74). Bangla has been argued to project a structural DP layer, within which the NP raises to Spec-DP, deriving structures of the type shown in (74) below (Bhattacharya 1999; Dayal 2012). Indefinite expressions, by contrast, are those in which this NP movement does not take place.

¹¹Owusu (2022) mentions the proximal determiner *yí* in Akan, which appears to have a similar distribution to *no*. I do not address it here, pending further empirical clarification. See Owusu (2022) and Owusu (2025) for discussion.

- (72) *chhatro du to ashe ni*
 student two CL come NEG
 ‘The two students didn’t come.’

- (73) *du to chhatro ashe ni*
 two CL student come NEG
 ‘Two students haven’t come.’



As in Akan, NP–CL constructions in Bangla are infelicitous with globally unique objects (75). However, Yip et al. (2023) show that this follows from bare NPs in Bangla being quasi-names (as was explained for Akan in the previous section). As expected of definite descriptions, NP–CL constructions in Bangla do not license contrastive readings (76a), whereas demonstratives, as in (76b), do.

- (75) *prodhan montri (#ta) aj kolkata-e ashchen*
 prime minister CL today kolkata-LOC come
 ‘The Prime Minister is coming to Kolkata today.’

- (76) a. *#kukur ta ghumochhe, kukur ta dourochhe*
 dog CL sleeping, dog CL running
 Intended: ‘This dog is sleeping, this dog is running.’
 b. *ei kukur ta ghumochhe, ei kukur ta dourochhe*
 DEM-PROX dog CL sleeping, DEM-PROX dog CL running
 ‘This dog is sleeping, this dog is running.’

Bangla has two demonstrative morphemes, *ei* (proximal) and *oi* (distal). As Dayal (2012) observes, demonstratives in Bangla appear in two distinct configurations as in (77a) and (77b) below. In the absence of the demonstrative morpheme *ei*, only (77b) is a definite description, while (77a) is an indefinite. However, both behave as a demonstrative construction.

- (77) a. *ei du to boi*
 DEM-PROX two CL book
 'These two books.'
- b. *ei boi du to*
 DEM-PROX book two CL
 'These two books'

Bhattacharya (1999) argues that while (77a) has a deictic reading, (77b) yields a specific interpretation. In practice, however, these configurations are difficult to distinguish in purely deictic contexts, as they are largely interchangeable. It is only in anaphoric environments that their contrast becomes visible. In (78) below, the configuration Dem-numeral-CL-NP (78b) is markedly more degraded.

- (78) *Diya bought two books and three paintings at yesterday's bookfair...*
- a. *oi boi du-to o onek din dhore kinte chaichhilo*
 DEM-DIST book two-CL 3.SG many day over buy want
 'She had wanted to buy those two books for a very long time now.'
- b. *#/?oi du-to boi o onek din dhore kinte chaichhilo*
 DEM-DIST two-CL book 3.SG many day over buy want

Intended: 'She had wanted to buy those two books for a very long time now.'

Building on Dayal's (2012) proposal, I argue that the deictic morphemes *oi* and *ei* in Bangla encode an iota operator, however, in light of the contrast in (78), I propose that they do not have an index argument. Instead, they contribute a relational property (which can be deixis, and must be deixis if supplemented with co-speech pointing) together with an anti-uniqueness presupposition, as in (79) below. This analysis correctly predicts that both configurations in (77a) and (77b) are available for deictic reference in the utterance context, while explaining why the configuration in (78b) is degraded under anaphoric use. Note that I am following Elbourne (2008) here in keeping the relational property not specified to a deictic one, and I will shortly address the motivation for it in the following.

(79) Definition of demonstrative determiners in Bangla

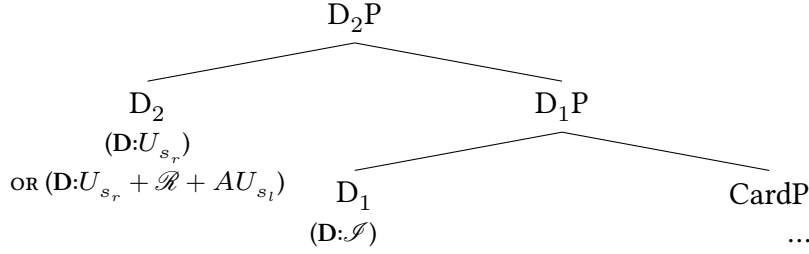
$$\llbracket \text{Dem}_{ei/oi} \rrbracket^{c,g} = \lambda P \lambda s. \iota x [P(x)(s) \wedge \mathcal{R}(x, \mathcal{S})(s)]$$

if $\exists! x [P(x)(s) \wedge \exists s' s \leq s' |P(x)(s')| > 1 \wedge \mathcal{R}(x, \mathcal{S})(s)]$, undefined otherwise.

where $\mathcal{R}$ is a relational property and $\mathcal{S}$ is the speaker

On this basis, I propose a split-D analysis for Bangla, illustrated in (80) below. The determiner that triggers NP movement is a D-morpheme hosted in D_1 , which introduces an index. The index needs to be bound by an ι operator, which D_2 does. The determiner D_2 is the locus of iota: it is realized either as a covert iota in NP-CL constructions or as the overt demonstrative *oi/ei*, which additionally encodes a deictic property and an anti-uniqueness presupposition. Placed within the (de)-compositional framework, D_1 hosts the $D:\mathcal{S}$ block, and D_2 can host either $D:U_{s_r}$ or $D:U_{s_r} + \mathcal{R} + AU_{s_l}$.

(80)



When D_1 projects, it triggers NP movement to Spec- D_1 , while D_2 by itself does not. If the referring expression has to be used anaphorically, the inversion is necessary, irrespective of whether D_2 hosts the covert iota operator, or the overt demonstratives *ei* and *oi*. There are three points to note regarding the relational property $\mathcal{R}$ in the demonstrative *oi* in Bangla. First, when D_1 is not projected, *oi* is not ambiguous: in the absence of an index, its only felicitous use involves co-speech gesturing, and the relational property is the deictic property DIST (for distal), specifically that the referent is located at a distance, as illustrated shortly in (85). Second, when D_1 is projected, an apparent ambiguity arises (similar to English; see §4.1) regarding the value of the relational property: it may either be the deictic property DIST or be trivially satisfied. Third, this ambiguity disappears once the demonstrative is supplemented with co-speech gesturing—even when D_1 is projected—since evaluation is then forced to be resolved in the utterance context, and the only felicitous relational property is the deictic property DIST. With respect to the demonstrative *ei*, similar patterns hold between English *this* and Bangla *ei*. The most natural interpretation is one in which the relational property is the deictic property PROX (for proximal), especially when supplemented with co-speech gesturing. As is the case for English *this*, Bangla *ei* can also plausibly be used in a non-deictic manner, but I leave a fuller investigation of this possibility to future work.

The derivations of NP-CL, Dem-numeral-CL-NP, and Dem-NP-CL constructions are given in (83-86). For ease of exposition, I take a few simplifying liberties. First, I assume a simplified (short-form) implementation of the cardinality operator Card. I adopt the semantics for Card proposed by Ionin and Matushansky (2006), given in (81) below. Put simply, Card is a function that introduces a partition over the domain of individuals creating a set that only includes n-sum individuals. For example, the Card 2 in *two boys* would take a set containing atomic individuals who have the property BOY and create a partition wherein each partition is comprised of exactly 2 boys and each of these boys can be predicated with another property - for example PLAYING as in *two boys are playing*.

(81) **Definition of n_{card}**

- a. $\llbracket n_{card} \rrbracket = \lambda P \lambda x. \exists S [\Pi(S)(x) \wedge |S| = n \wedge \forall s \in S P(s)]$
- b. $\Pi(S)(x) = 1$ iff S is a cover of x , and $\forall z, y \in S [z=y \vee \neg \exists a [a \leq z \wedge a \leq y]]$ (Forbidding that cells of the partition overlap ensures that no element is counted twice.)
- c. A set of individuals C is a cover of a plural individual X iff X is the sum of all members of
of
 $C: \sqcup C = X$

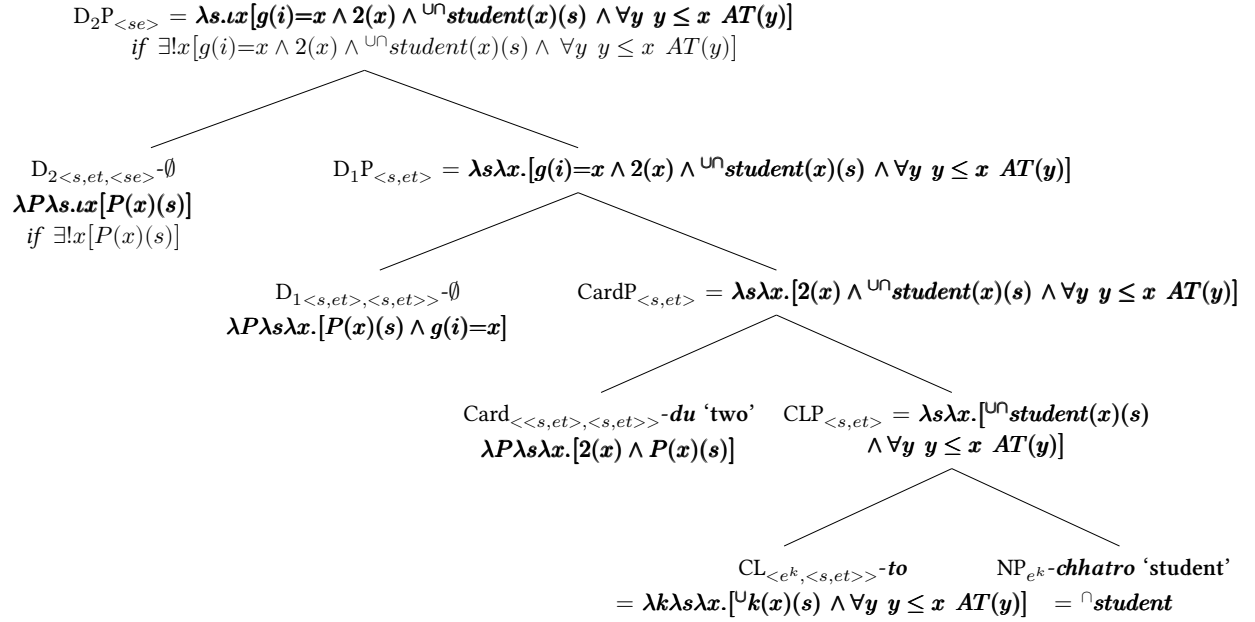
(Ionin and Matushansky 2006)

In the following derivations, I will use the numeral to represent this function, as in (82). This is working as a placeholder, and if we plug in the definition in (81), the logic should remain the same.

$$(82) \quad \llbracket two \rrbracket = \lambda P. \lambda x [2(x) \wedge P(x)]$$

Second, I have kept the NP at its base position in the composition, but it can certainly be interpreted higher in Spec-D₁ via Predicate Abstraction between Spec-D₁ and D₁.

(83) Deriving *chhatro du-to* ‘student two-CL’¹² (NP movement triggered by the projection of D₁)

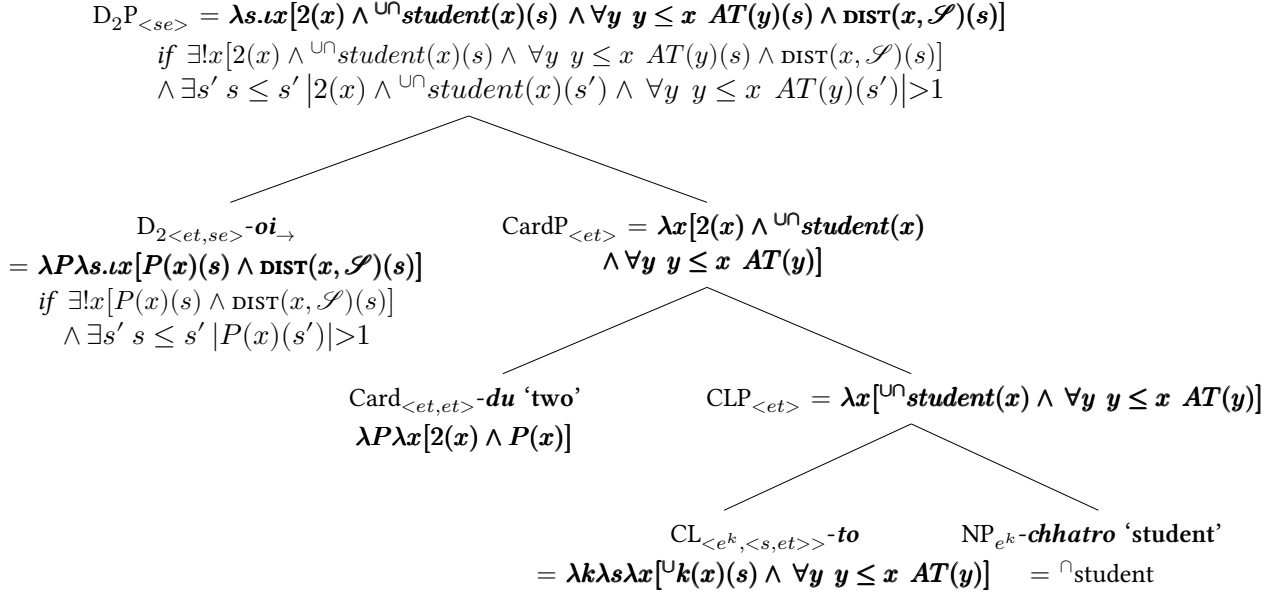


The unique two-sum consisting two students, defined if there is exactly one plural individual which is a two-sum consisting two students in a relevant situation *s*

¹²Given that Bangla is a classifier language that lacks systematic number marking, I assume, following Chierchia (1998), that the noun enters the lexicon as a kind individual of type e^k . Thus, in (83–86), the interpretation of the NP *chhatro* ‘student’ is represented using the $\bigcap$ (down) operator, as proposed in Chierchia (1982) and Chierchia (1998). This operator corresponds to the type-shifting operation *nom*, which shifts an ordinary predicate to a kind term.

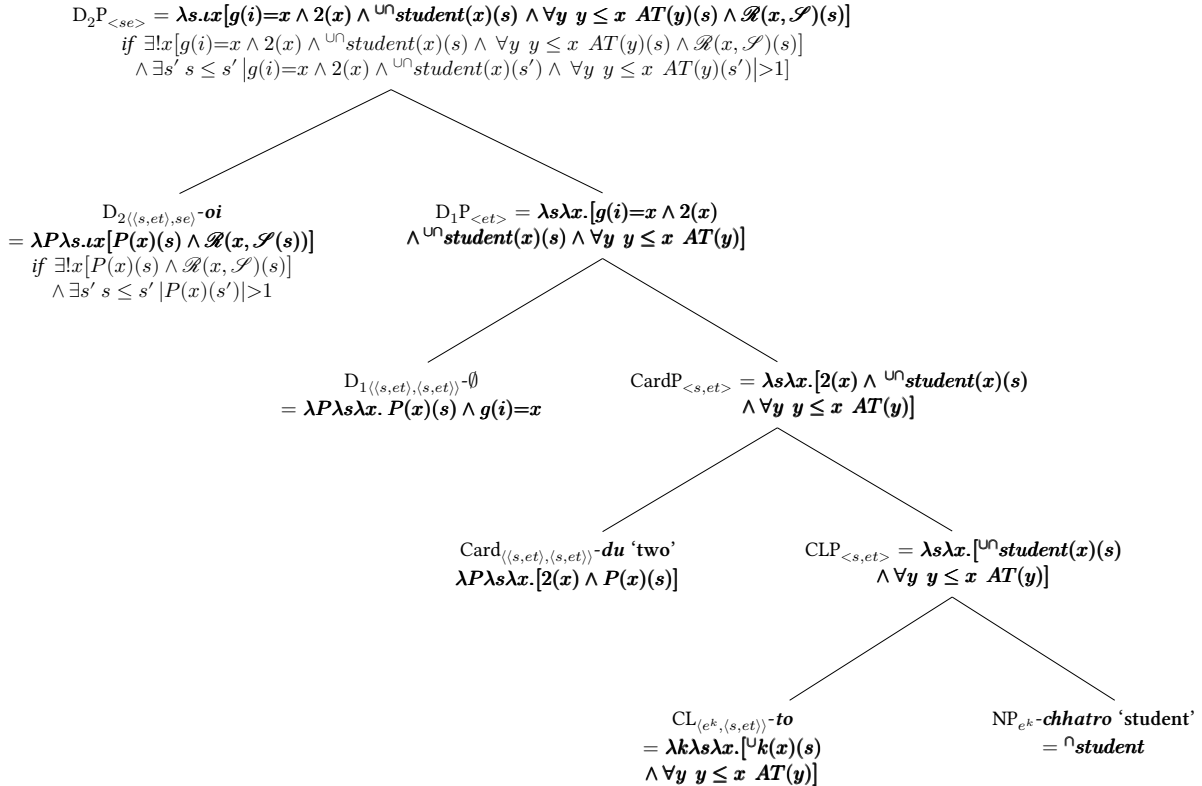
(84) *nom*: $P \rightarrow \bigcap P$

(85) Deriving *oi* → *du-to chhatro* ‘Dem two-cl student’



‘The unique two-sum consisting two students who are located at a distance from the speaker in the utterance context, defined if there is exactly one plural individual which is a two-sum consisting two students who are located at a distance from the speaker in the utterance context.’

(86) Deriving *oi chhatro du-to* ‘Dem student two-cl’ (NP movement triggered by the projection of D_1)



‘The unique two-sum consisting two students who are located at a distance from the speaker in the utterance context, defined if there is exactly one plural individual which is a two-sum consisting two students who have a relevant relational property’¹³ in the utterance context.’

In (86) above, the demonstrative is interpreted as a unique plural individual in the restricted situation s , so long as there is a larger situation with respect to s , where the exact unique plural individual is not unique - there are other two-sums of students which could potentially be assigned different indices and deictic properties.

5 Concluding remarks

Across Mizo, Akan, and Bangla, demonstrative systems converge on a shared division of labor between reference, anti-uniqueness, and indexicality, despite surface differences in morphology. In Mizo, deictic determiners introduce an iota operator together with a deictic presupposition, while anaphoric determiners introduce an iota operator and an index. By contrast, the demonstrative functions as an NP modifier and contributes an anti-uniqueness presupposition. A parallel pattern is found in Akan: the determiner *nó* is a D-head that introduces an iota operator and an index, whereas *saa* is an NP modifier that contributes an anti-uniqueness presupposition. Thus, both Mizo and Akan instantiate a system in which anti-uniqueness is specifically encoded by an NP-modifier. Unlike Akan and Mizo, Bangla encodes anti-uniqueness in the same morpheme that also introduces an iota operator and a deictic property. I have proposed a split-D analysis for Bangla, wherein there are two syntactic positions where D-morphemes can be hosted. One of them, D1, introduces just an index. An idiosyncratic property of D1 is that it triggers NP movement to Spec of D1. The other D head, D2, can have two instantiations: one that is simply a covert iota operator, and another which additionally introduces an anti-uniqueness as well as a deictic presupposition.

Taken together, the three languages show that demonstratives are composite structures, minimally encoding an anti-uniqueness presupposition and immediate situation uniqueness. The observed variation across languages reduces to how these components are structurally distributed within the DP spine. What the crosslinguistic picture reveals, more broadly, is that the traditional opposition between definites and demonstratives is not a primitive distinction but an emergent one: it reflects different combinations of independently available reference-building mechanisms, assembled differently across languages. The four components identified here — USr , $\mathcal{I}$, $\mathcal{R}$, and $AUSl$ — are not tied to any particular morphological form, and languages vary freely in how they package and distribute them.

The Mizo data are particularly illuminating in this regard, in that they render transparent a decomposition that remains morphologically opaque in languages like English. The tentative extension to the *this/that* distinction in English, explored in §4.1, suggests that even in a language where D-expression properties are fused within a single exponent, the underlying components may still leave distributional traces — in particular, in the differential availability of anaphoric readings and the sensitivity to co-speech pointing. Whether this parallelism holds up under closer scrutiny is left for future work, but the Mizo facts at minimum provide a new vantage point from which to approach these questions.

¹³When supplemented with co-speech pointing, the relational property specifies that the two-sum stands in a distal relation to the speaker. In the absence of co-speech gesturing, the expression receives a purely anaphoric interpretation, and the relational property is trivially satisfied.

Language	Category/Element	D-features	Iota	Index	Deictic	Anti-unique
Mizo	Deictic definite	$D:U_{s_r} + \mathcal{R}$	✓	–	✓	–
	Anaphoric definite	$D:U_{s_r} + \mathcal{I}$	✓	✓	–	–
	Demonstrative	$D:AU_{s_l}$	–	–	–	✓
Akan	Definite (<i>nó</i>)	$D:U_{s_r} + \mathcal{I}$	✓	✓	–	–
	Demonstrative (<i>saa</i>)	$D:AU_{s_l}$	–	–	–	✓
Bangla	D ₁ (covert)	$D:\mathcal{I}$	–	✓	–	–
	D ₂ (covert)	$D:U_{s_r}$	✓	–	–	–
	D ₂ (demonstrative)	$D:U_{s_r} + \mathcal{R} + AU_{s_l}$	✓	–	✓	✓

Table 3: Division of labor in demonstrative systems across Mizo, Akan, and Bangla

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